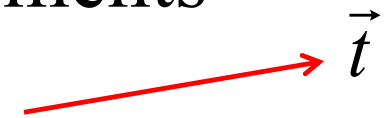
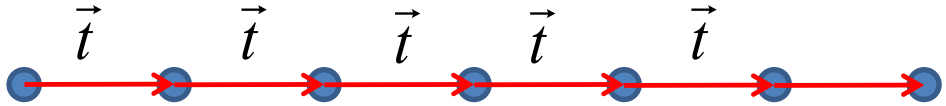
III Crystal Symmetry

3-1 Symmetry elements

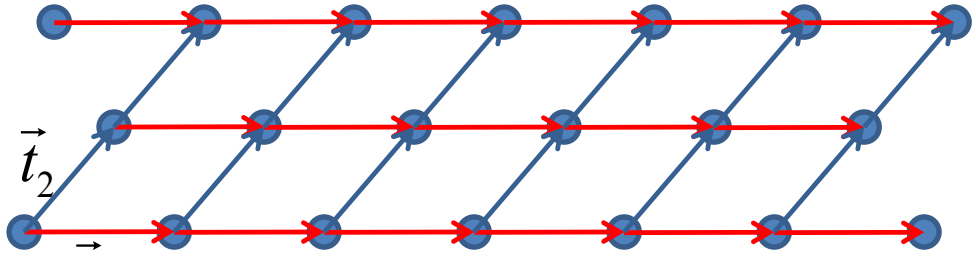
(1) Translation:



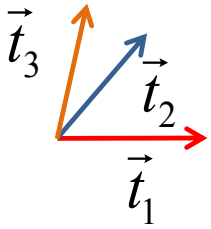
Example: 1D lattice:



2D lattice:



3D lattice:



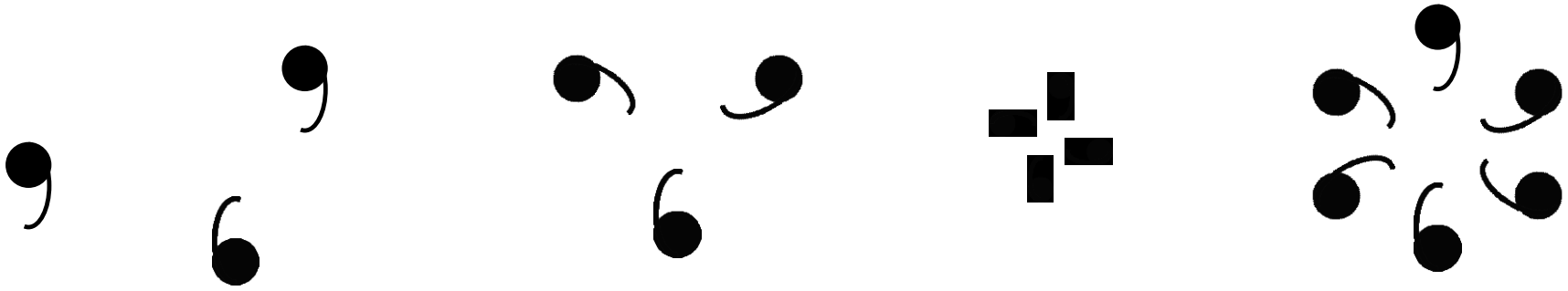
$$\vec{t} = u_1\vec{t}_1 + u_2\vec{t}_2 + u_3\vec{t}_3$$

Crystal = "motif" repeated at each lattice point

Atom, molecule

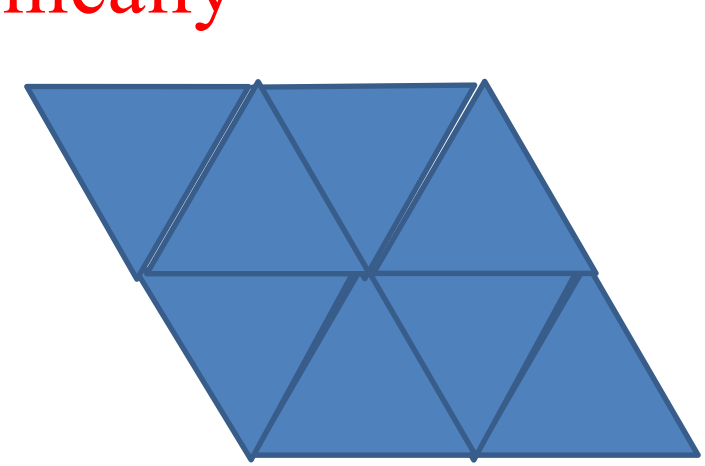
(2) Rotation symmetry

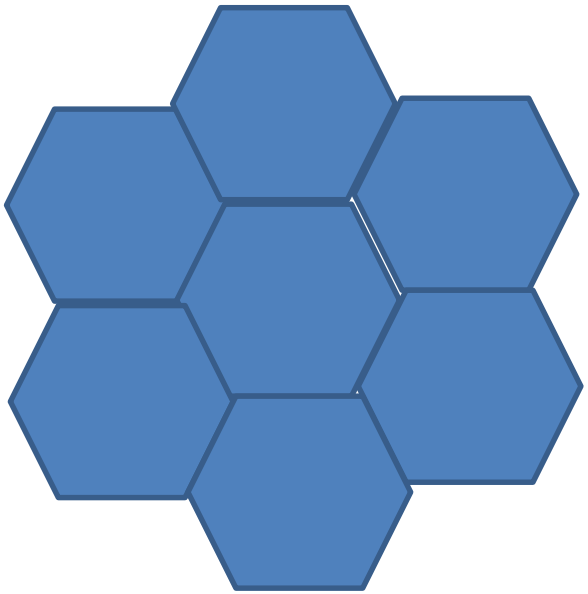
- one fold 1
- two fold (diad ⬮) 2
- three fold (triad ▲) 3
- Four fold (tetrad ■) 4
- Six fold (hexad ⬢) 6

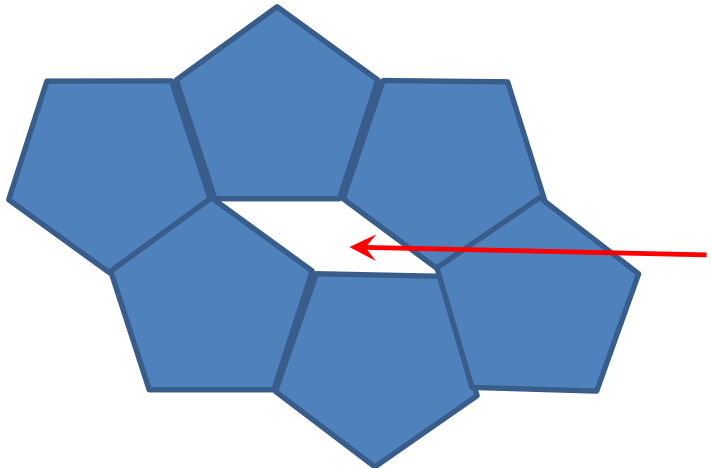


Proof: There are only 1, 2, 3, 4, and 6 fold rotation symmetries for crystal with translational symmetry.

graphically





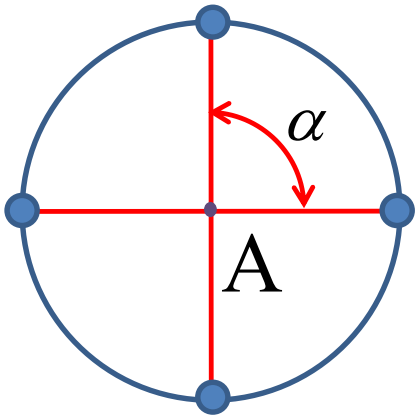


The space cannot be filled!

Axis A, angle α $\qquad \alpha = \frac{2\pi}{n}$ $\qquad n$: integer

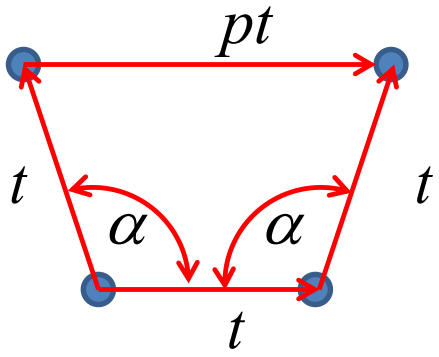
n fold axis $\qquad$ Notation n

Example:



4

Only certain types of rotations are consistent with lattice Periodicity (kepler):



$$t + 2t\sin(\alpha - \frac{\pi}{2}) = pt \qquad p\text{: integer:}$$

$$1 - 2\cos\alpha = p$$

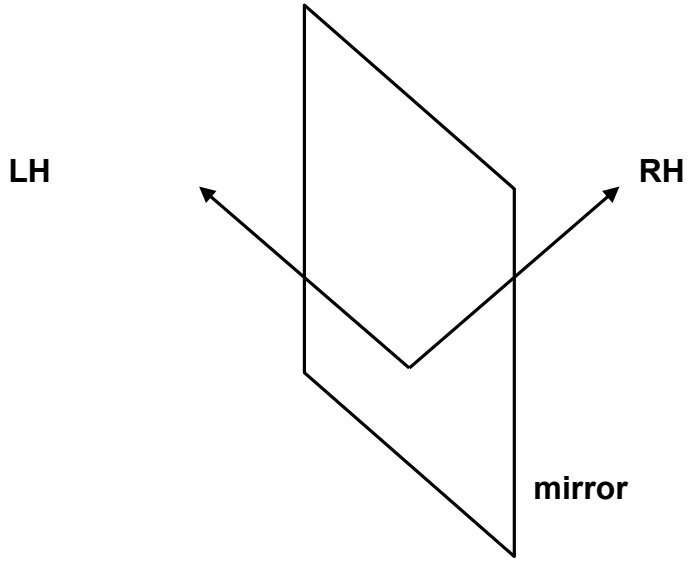
$$\boxed{\cos\alpha = \frac{1-p}{2}}$$

$$\cos\alpha = \frac{1-p}{2}$$

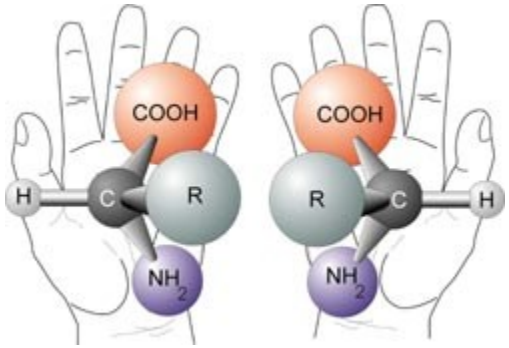
p	$\cos\alpha$	α	n
3	-1	180°	2
2	-1/2	120°	3
1	0	90°	4
0	1/2	60°	6
-1	1	0°	1

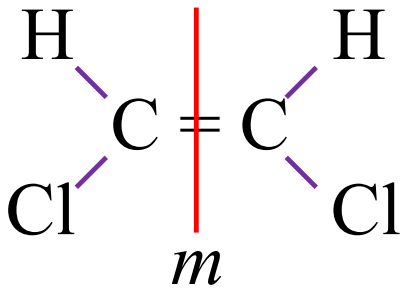
Crystallographic rotation axes

(3) Reflection (mirror) symmetry m

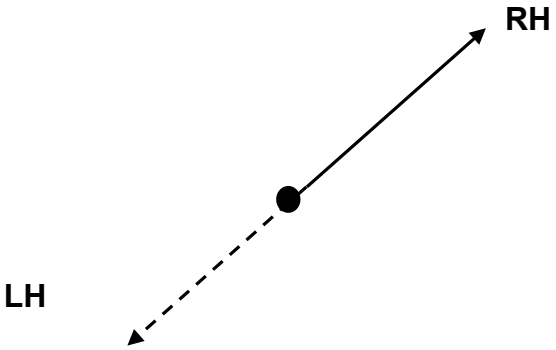


Examples:

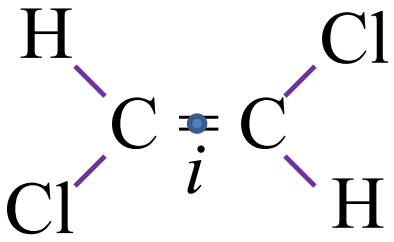




(4) Inversion symmetry (center of symmetry) $\bar{1}$



Example:



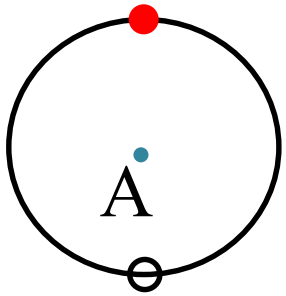
Combinations of Symmetry Elements

(5) Rotation-Inversion axis

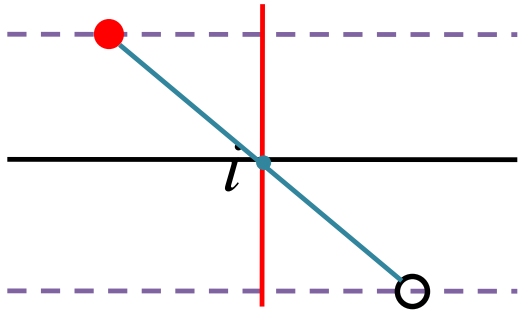
$= \text{Rotate by } \frac{360^{o}}{n} \cdot \text{ then invert.}$

(a) one fold rotation inversion $(\bar{1})$

top view

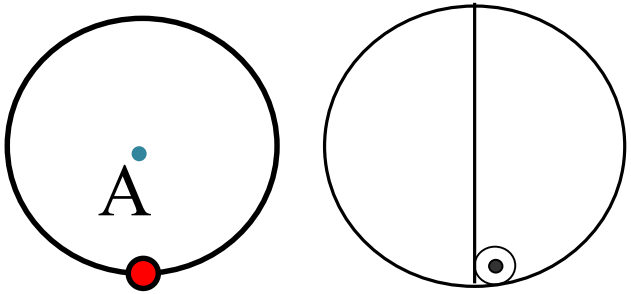


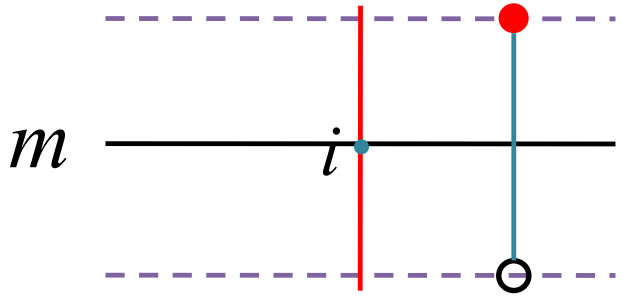
Side view



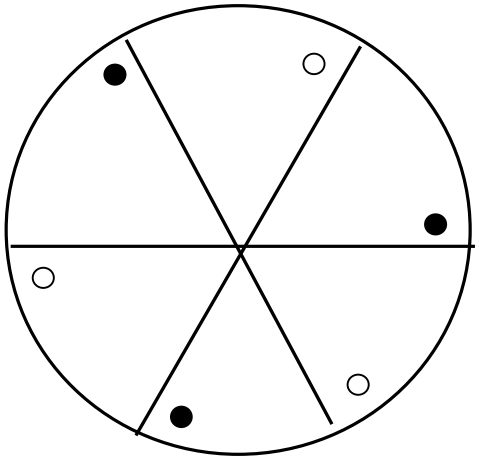
(b) two fold rotation inversion $(\bar{2})$

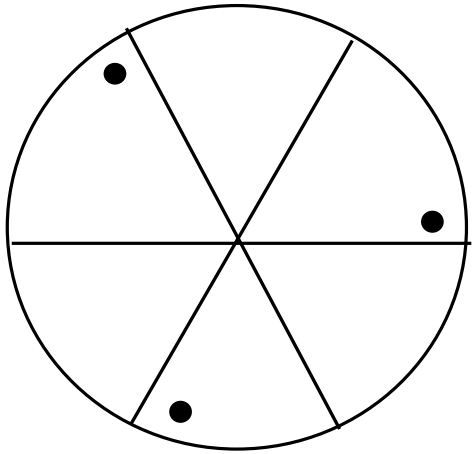
= mirror symmetry (m)



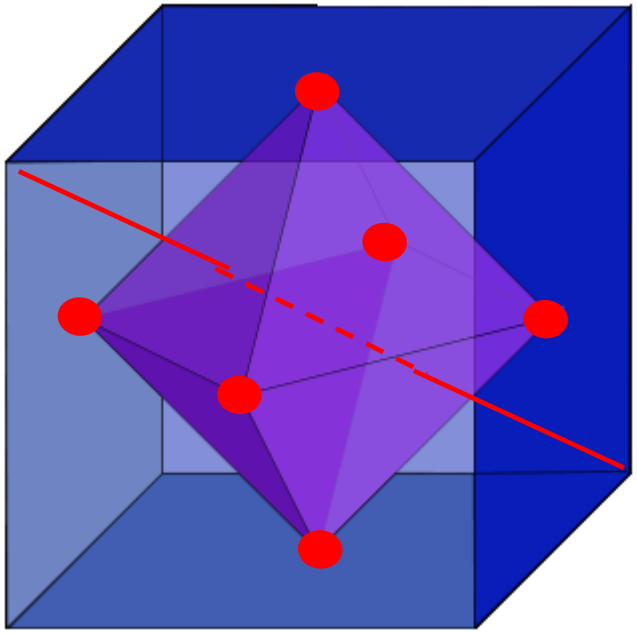


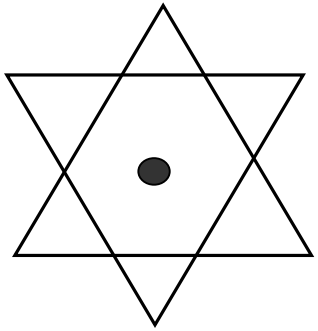
(c) inversion triad $(\bar{3})$ $\bar{3} = 3 \cdot i$





3

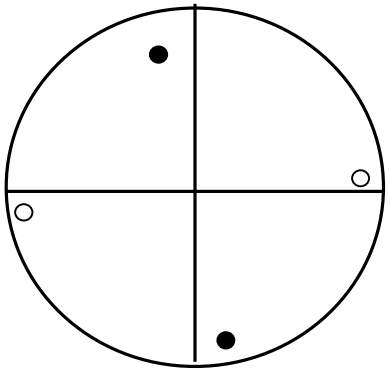


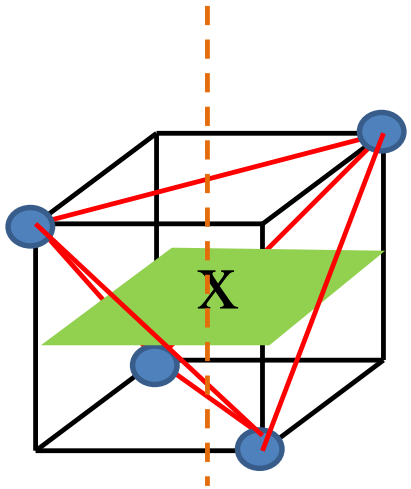


= Octahedral site in an octahedron

(d) inversion tetrad $(\bar{4})$

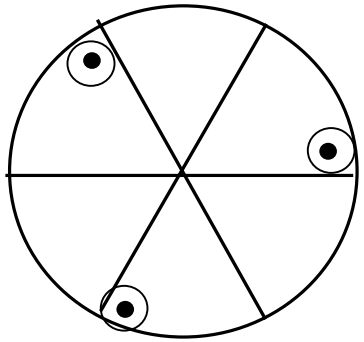
$\bar{4} \neq 4 \cdot i$



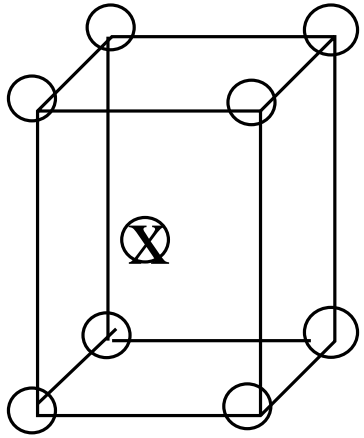


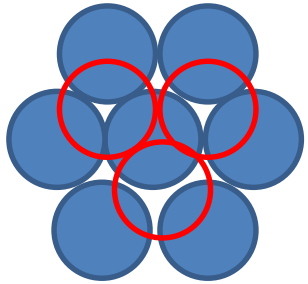
= tetrahedral site in a tetrahedron

(e) inversion hexad $(\bar{6})$ $\bar{6} \neq 6 \cdot i$



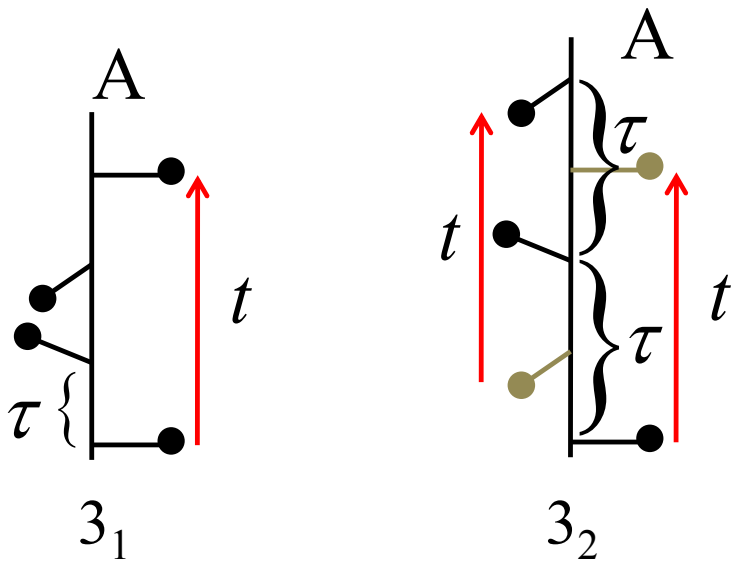
$$\bar{6} = \frac{3}{m}$$





Hexagonal close-packed (hcp) lattice

(6) Screw axis: rotation about A, then translation by τ // A

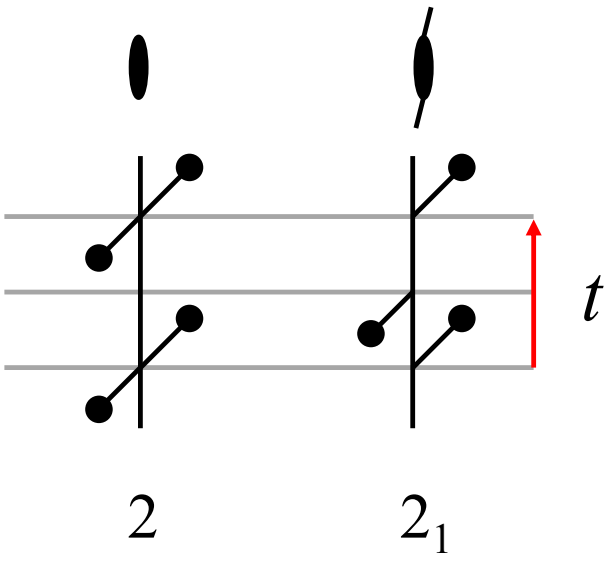


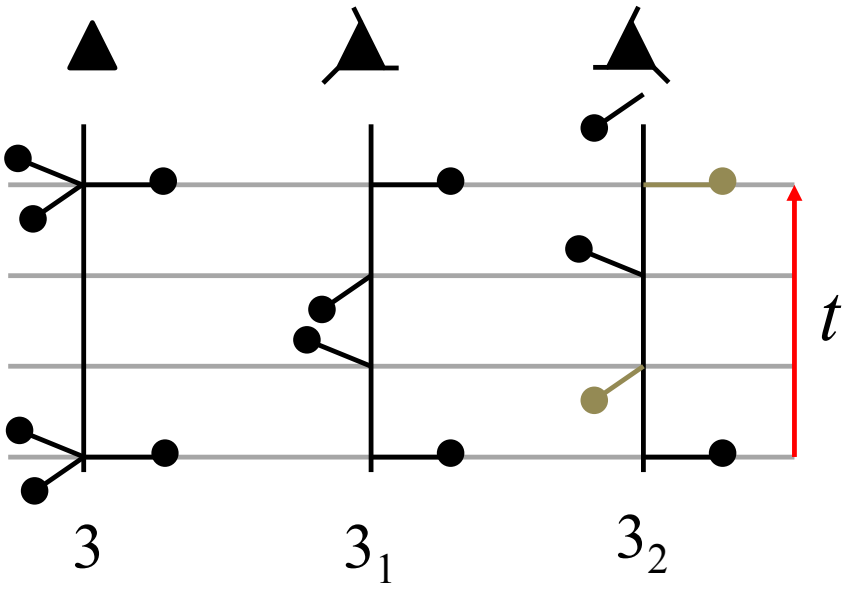
t: lattice translation vector along A

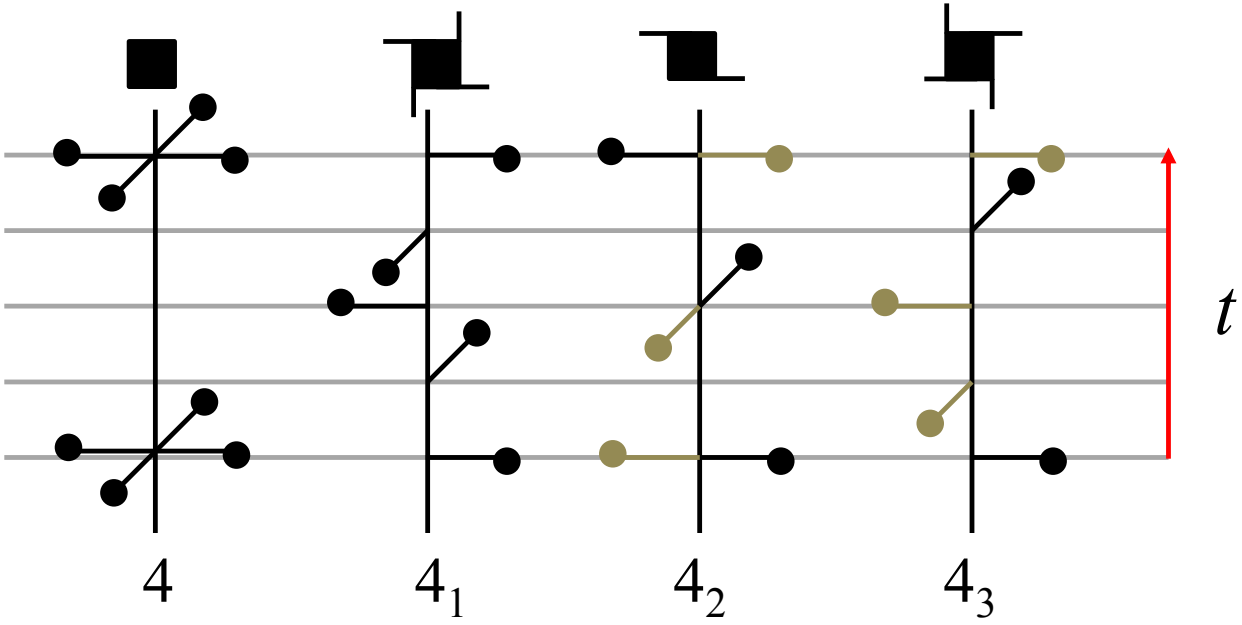
In a lattice: translation vector along A is *t*

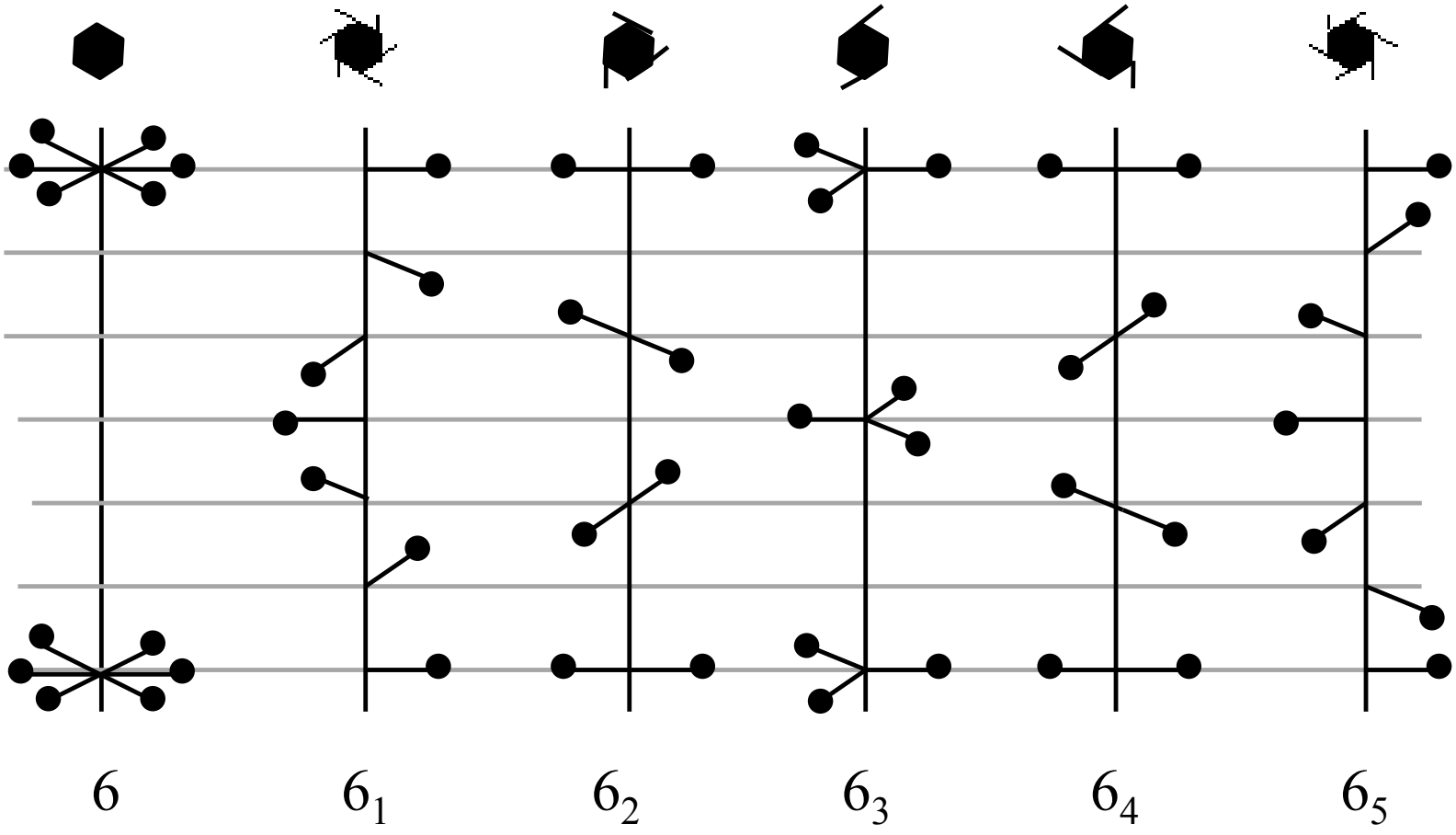
requirement: $\underbrace{n\tau}_{\text{pitch}} = mt \rightarrow \tau = \frac{m}{n}t$

for the "crystallographic" screw axes

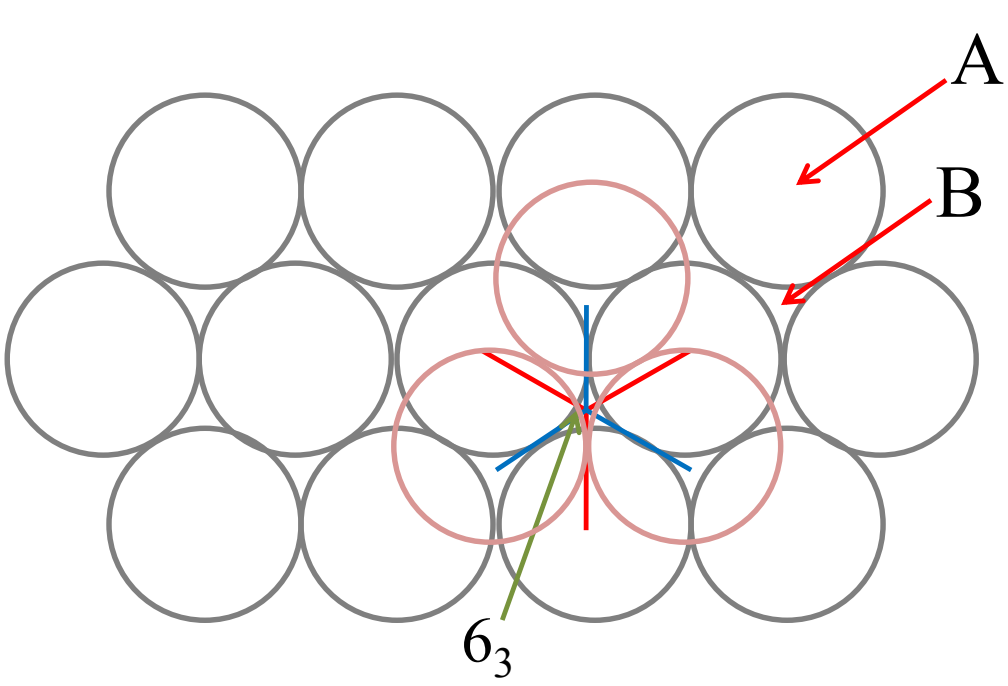

t
2
2₁

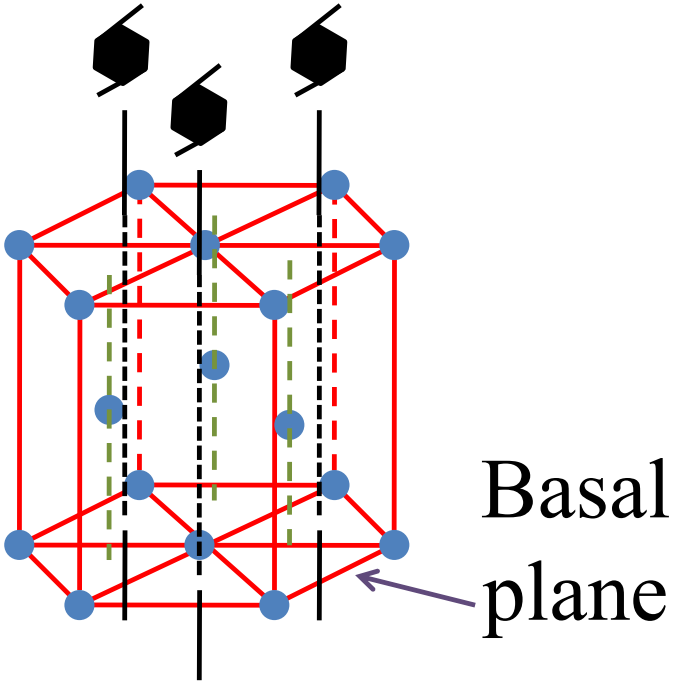

t
3
3₁
3₂


t
4
4₁
4₂
4₃

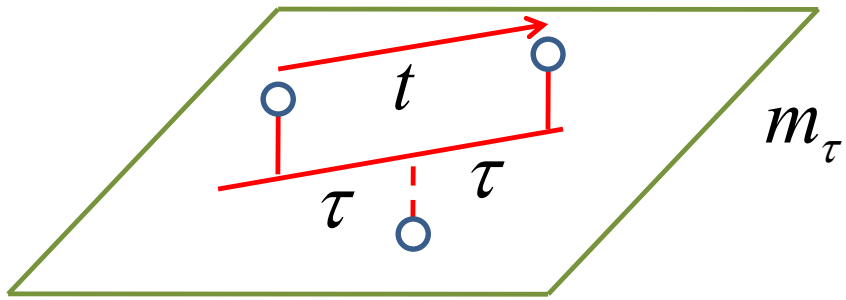

6
6_1
6_2
6_3
6_4
6_5

Example of 6_3 screw axis: hcp





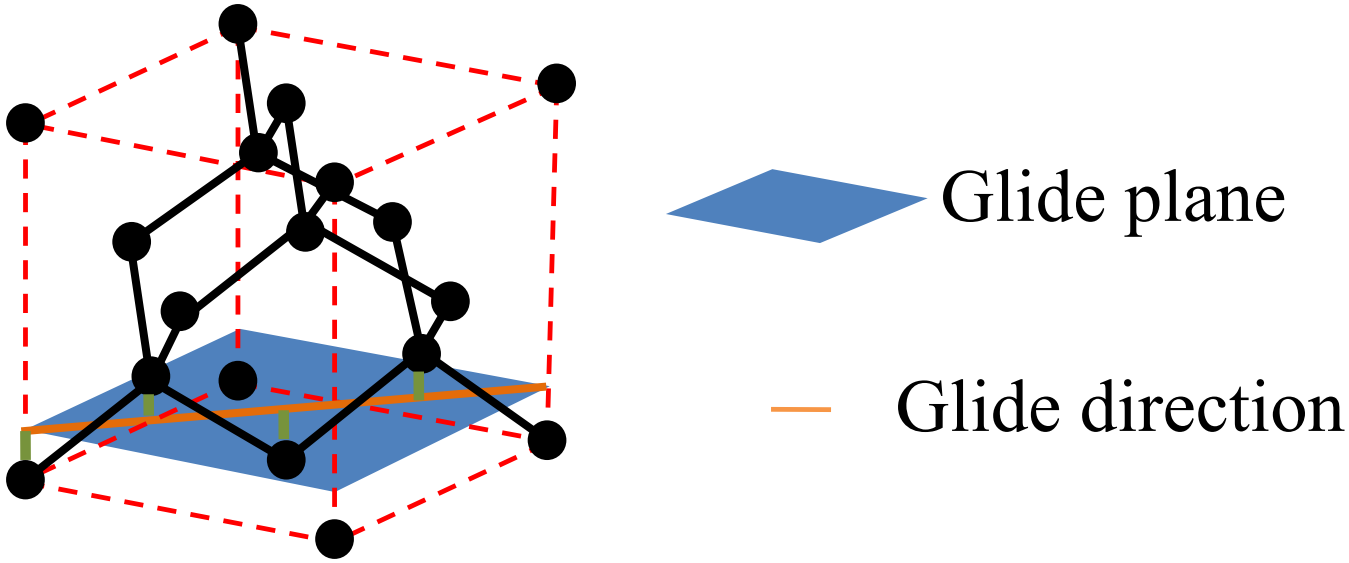
(7) Glide plane: translation by τ (in m), reflect about m



In lattice: t = translation distance along glide direction

$$\tau = t/2$$

Example: diamond cubic



Screw axis and glide plane involve translation: symmetry elements for space group not for point group!

Symmorphic space group is defined as a space group that may be specified entirely by symmetry operation acting at a common point (the operations need not involve τ) as well as the unit cell translation. (73 space groups)

Nonsymmorphic space group is defined as a space group involving at least a translation τ.

3-2. Fourteen Bravais lattice structures

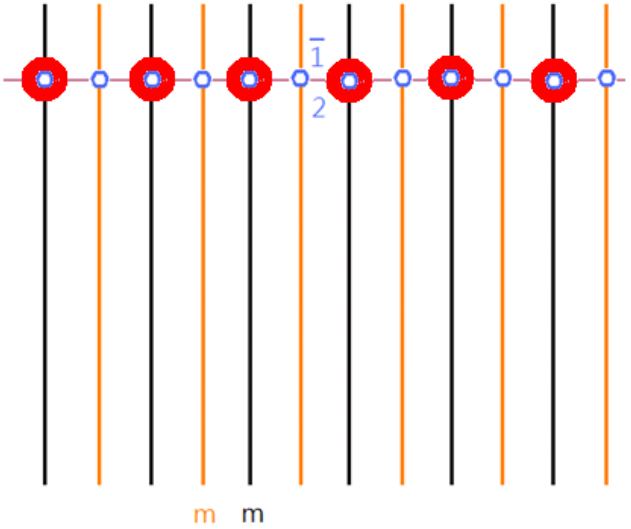
3-2-1. 1-D lattice

3 types of symmetry can be arranged in a 1-D lattice

mirror symmetry (m)

= 2-fold rotation (2)

= center of symmetry ($\bar{1}$)

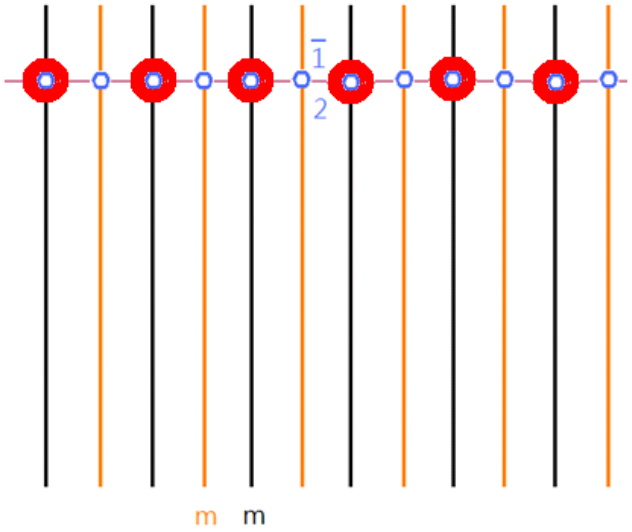


3-2-2. 2-D lattice

Two ways to repeat 1-D $\Rightarrow$ 2D

(1) maintain 1-D symmetry

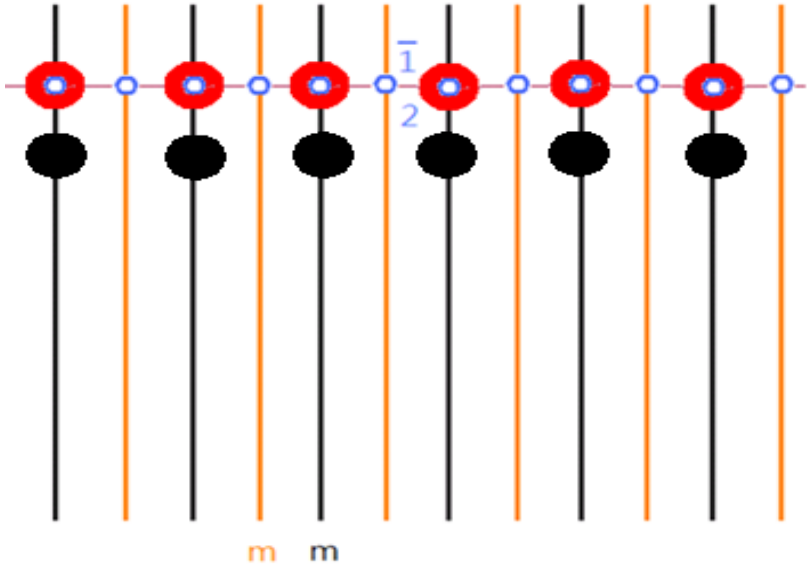
(2) destroy 1-D symmetry

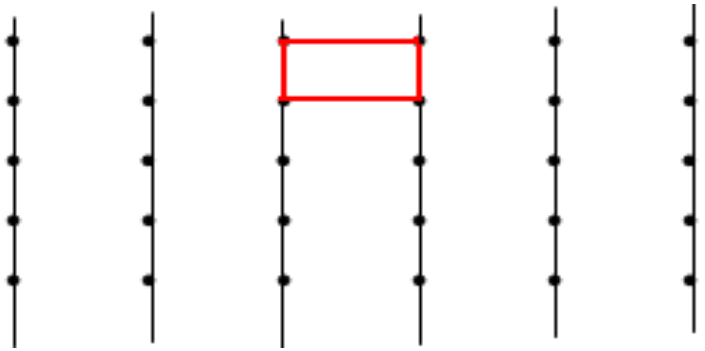


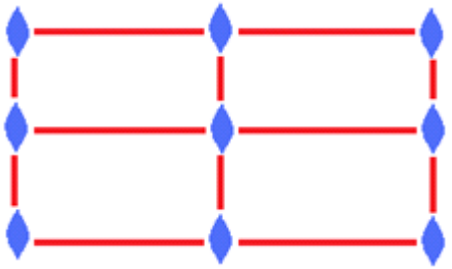
$m\ 2\ \bar{1}$

(a) Rectangular lattice ($a \neq b; \gamma = 90^{o}$)

Maintain mirror symmetry m



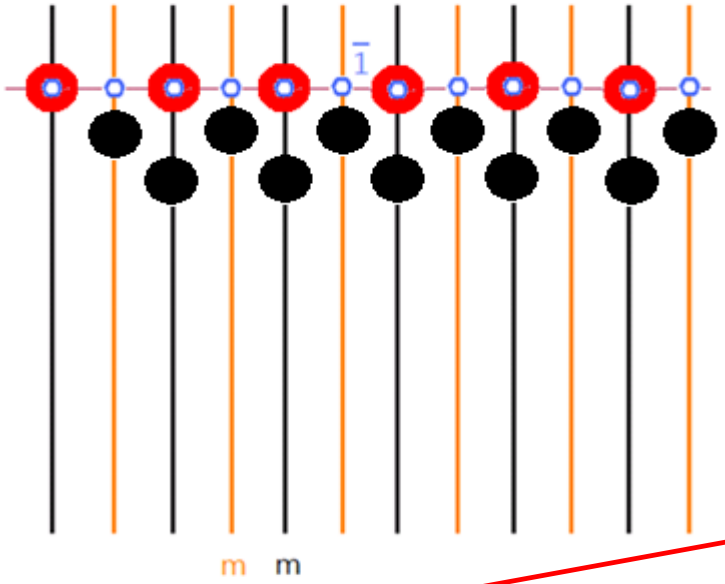


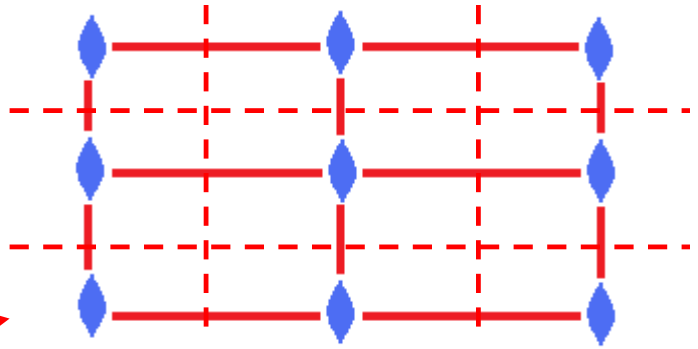


($a \neq b; \gamma = 90^{o}$)

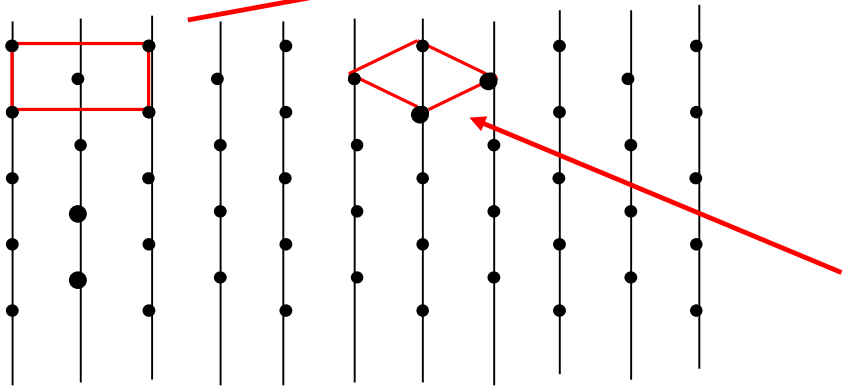
(b) Center Rectangular lattice ($a \neq b; \gamma = 90^o$)

Maintain mirror symmetry m





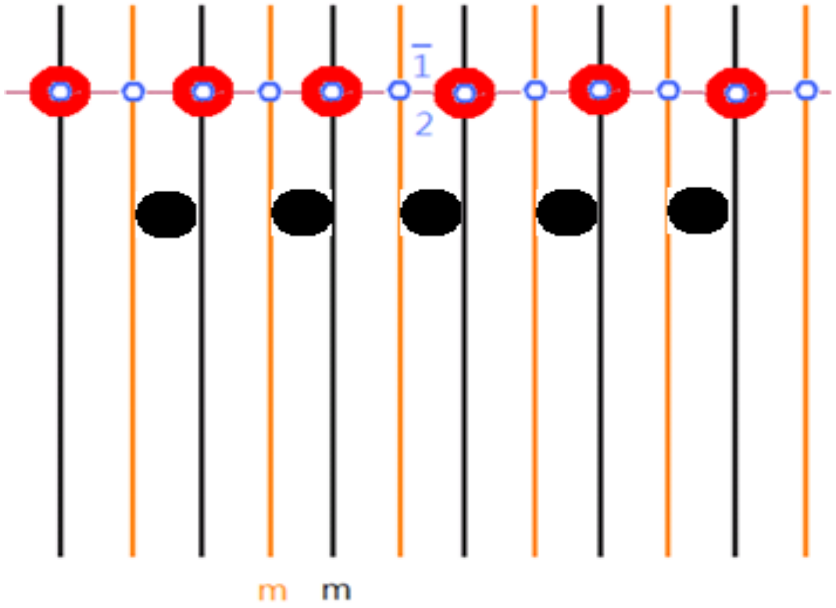
$(a \neq b; \gamma = 90^o)$

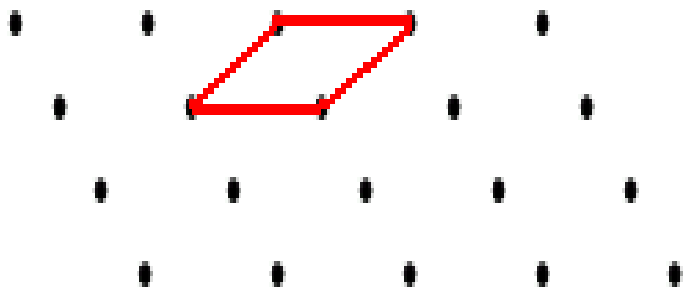


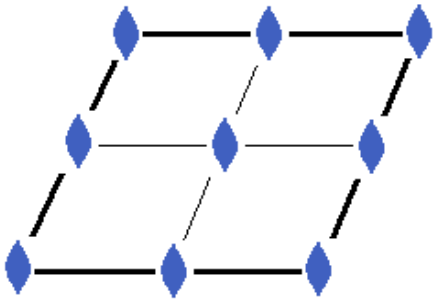
Rhombus cell
(Primitive unit cell)
$a = b; \gamma \neq 90^o$

(c) Parallelogram lattice ($a \neq b$; $\gamma \neq 90^{o}$)

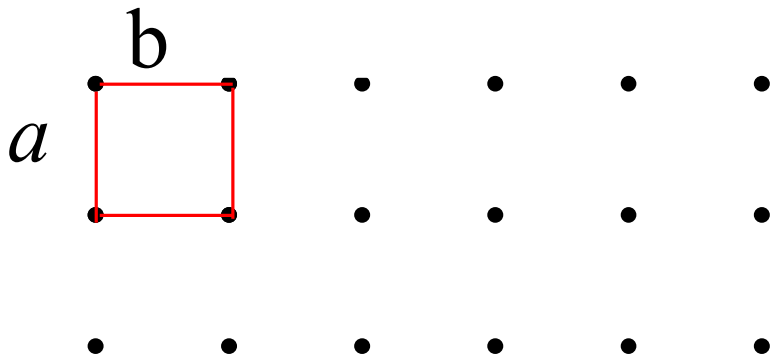
Destroy mirror symmetry

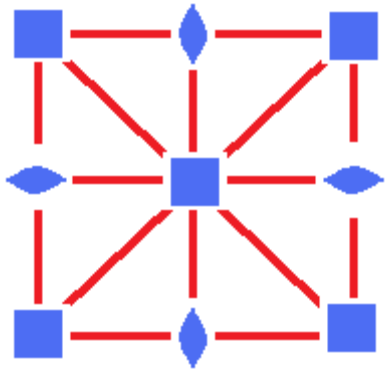




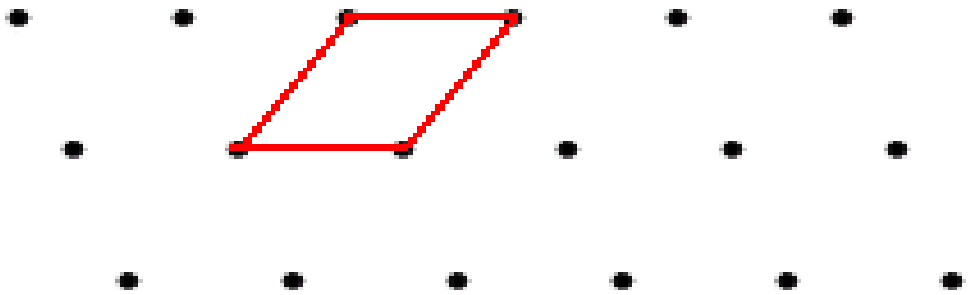


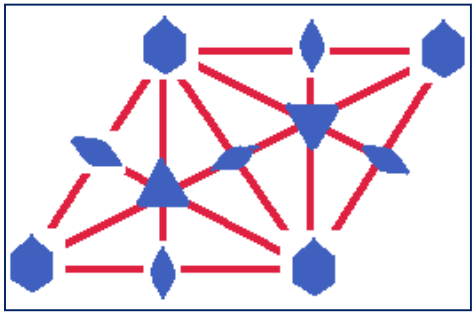
(d) Square lattice ($a = b; \gamma = 90^o$)



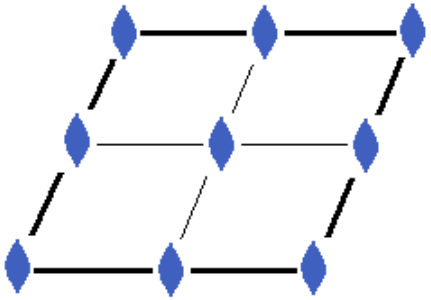


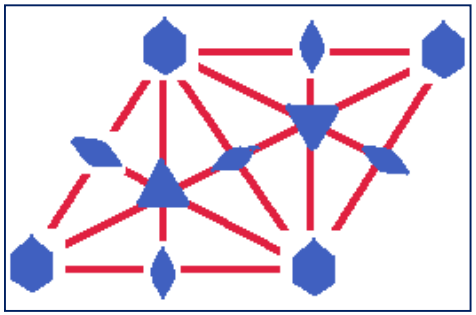
(e) hexagonal lattice ($a = b; \gamma = 120^o$)

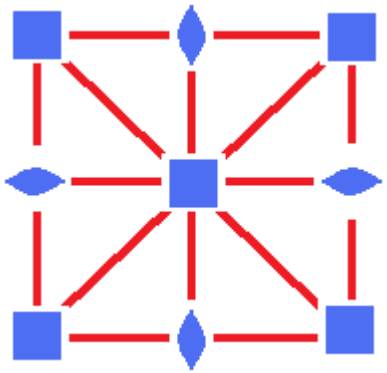


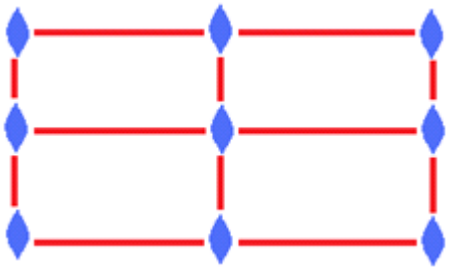


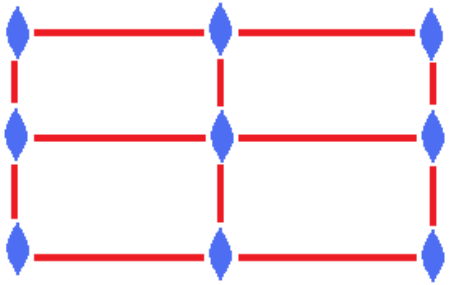
Symmetry elements in 2D lattice











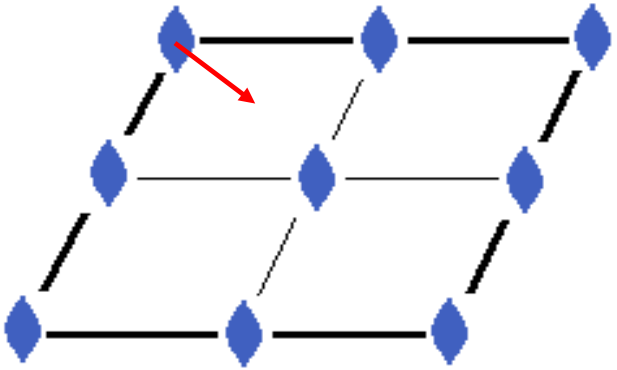
Rectangular = center rectangular?

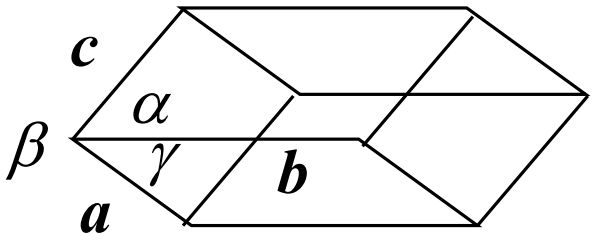
3-2-3. 3-D lattice: 7 systems, 14 Bravais lattices

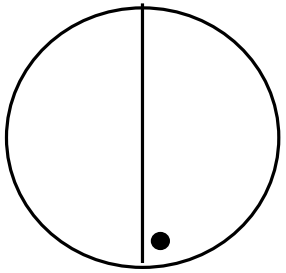
Starting from parallelogram lattice

$(a \neq b; \gamma \neq 90^{o})$

(1)Triclinic system 1-fold rotation (1)

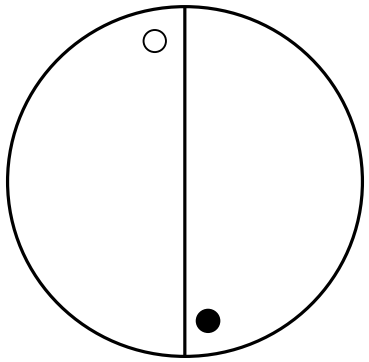




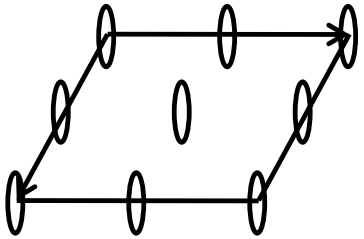


$(a \neq b \neq c; \alpha \neq \beta \neq \gamma \neq 90^{o})$

lattice center symmetry at lattice point as shown above which the molecule is isotropic ($\bar{1}$)



(2) Monoclinic system

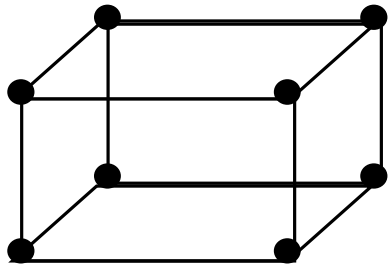


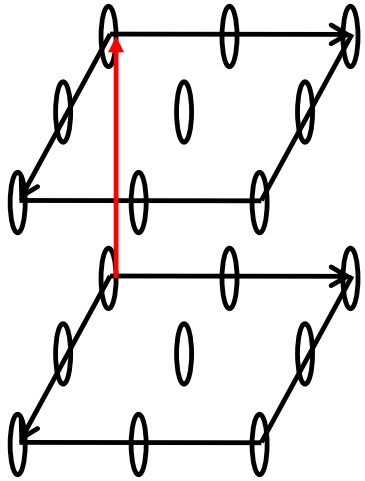
$(a \neq b \neq c; \alpha = \beta = 90^{o} \neq \gamma)$

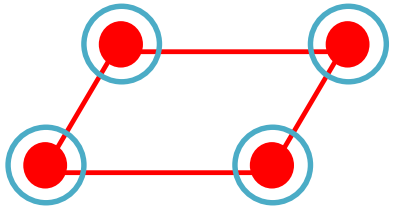
one diad axis

(only one axis perpendicular to the drawing plane maintain 2-fold symmetry in a parallelogram lattice)

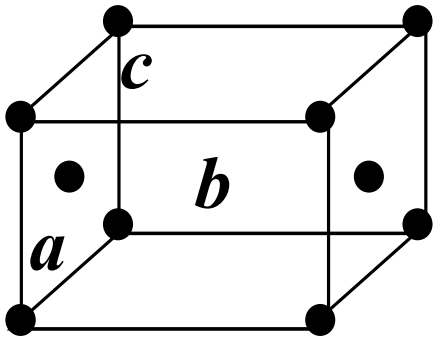
(1) Primitive monoclinic lattice (P cell)

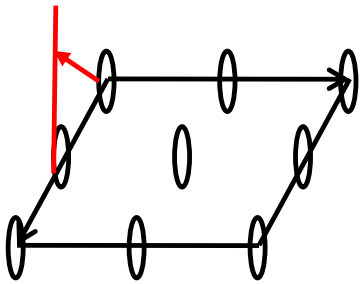


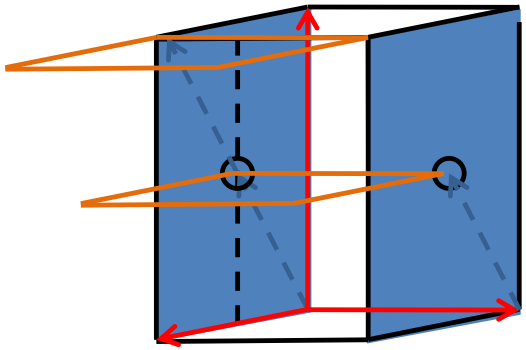




(2) Base centered monoclinic lattice



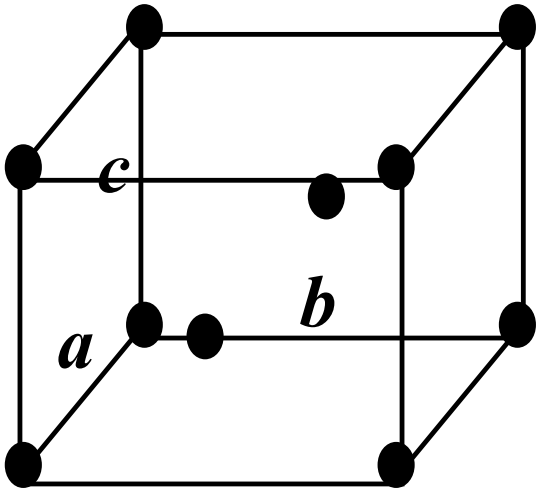


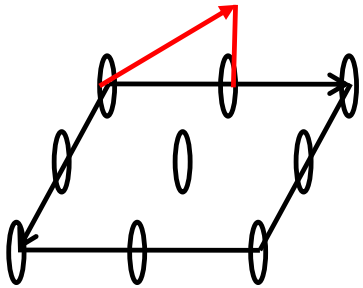


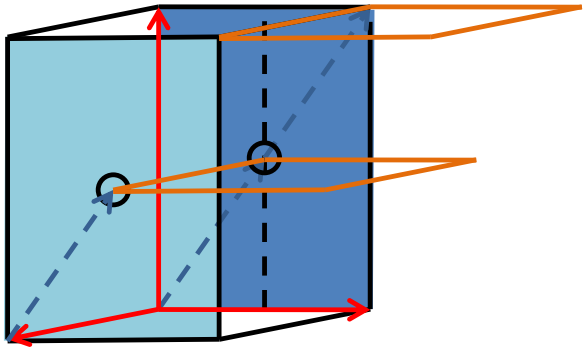
B-face centered monoclinic lattice

The second layer coincident to the middle of the first layer and maintain 2-fold symmetry

Note: other ways to maintain 2-fold symmetry



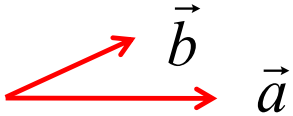




A-face centered monoclinic lattice

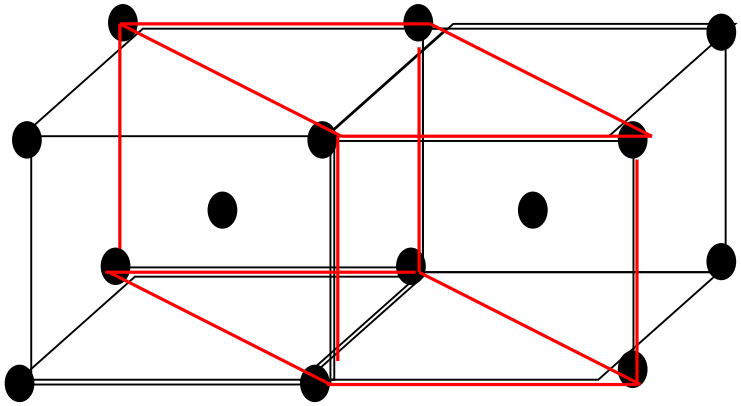
If relabeling lattice coordination

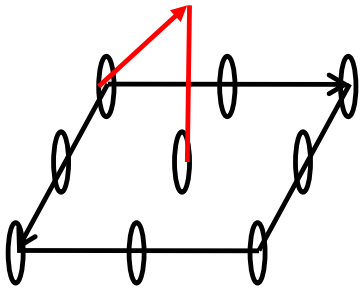


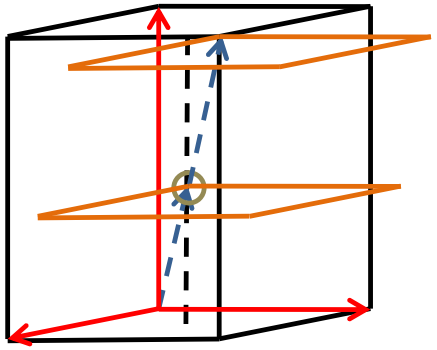


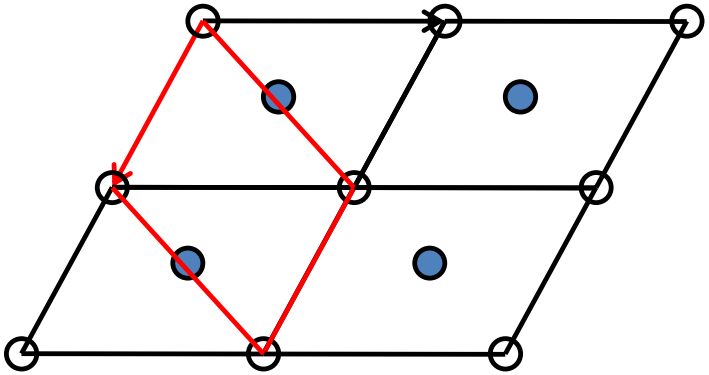
A-face centered monoclinic = B-face centered

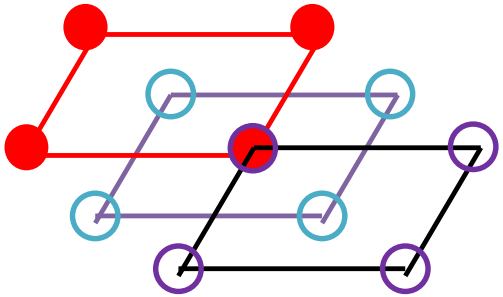
(2) Body centered monoclinic lattice









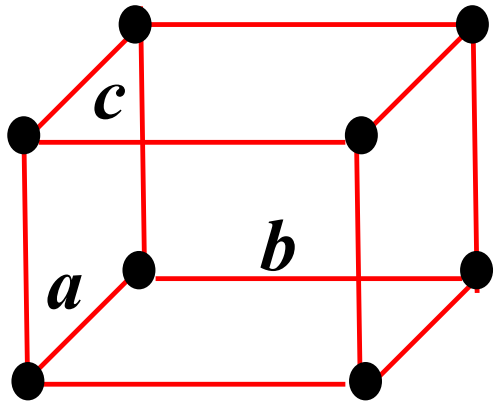


Body centered monoclinic = Base centered monoclinic

So monoclinic has two types

1. Primitive monoclinic
2. Base centered monoclinic

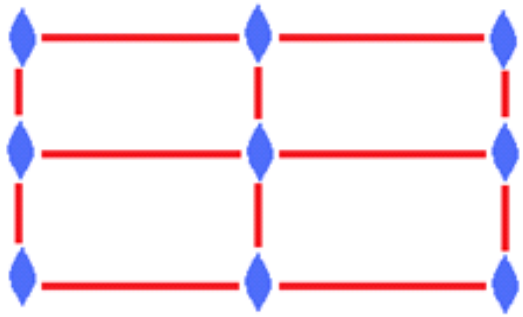
(3) Orthorhombic system



$(a \neq b \neq c; \alpha = \beta = \gamma = 90^{o})$

3 -diad axes

(1) Derived from rectangular lattice

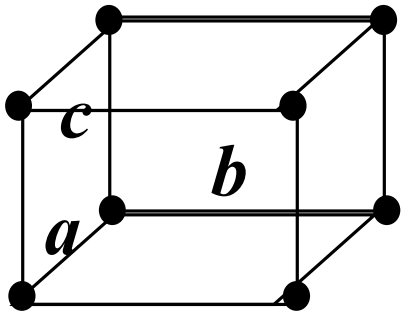


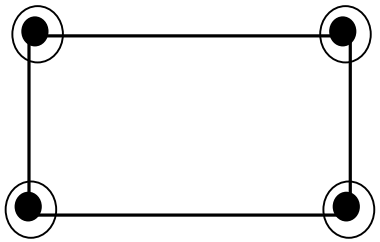
$(a \neq b; \gamma = 90^{o})$

→ to maintain 2 fold symmetry

The second layer superposes directly on the first layer

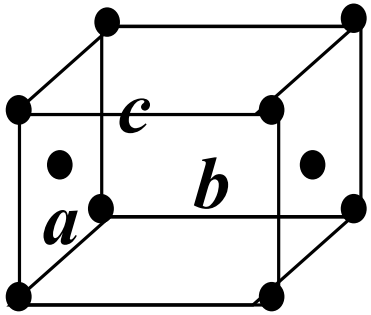
(a) Primitive orthorhombic lattice

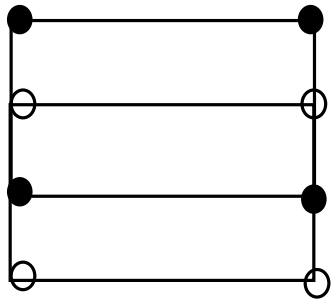


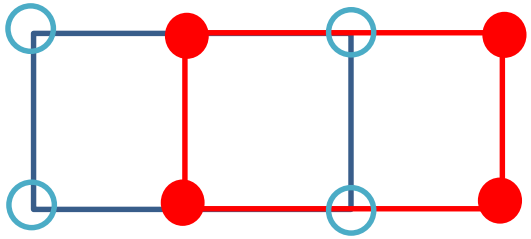


(b) B- face centered orthorhombic

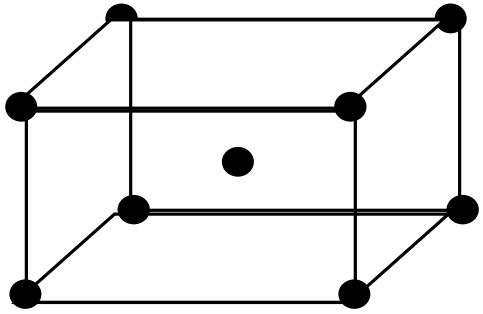
= A -face centered orthorhombic

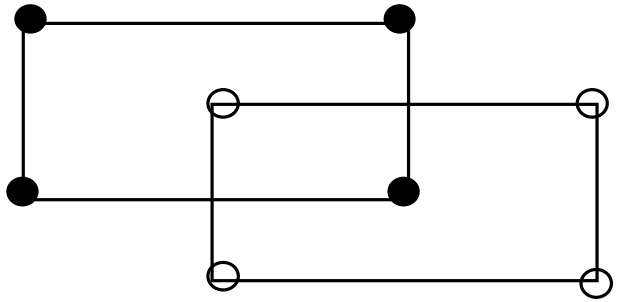


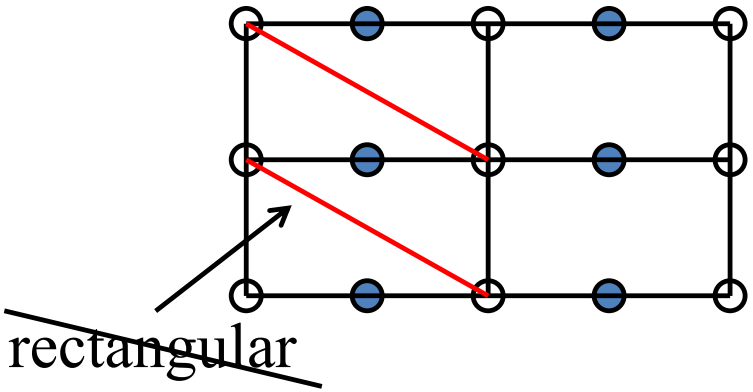




(c) Body-centered orthorhombic (I- cell)



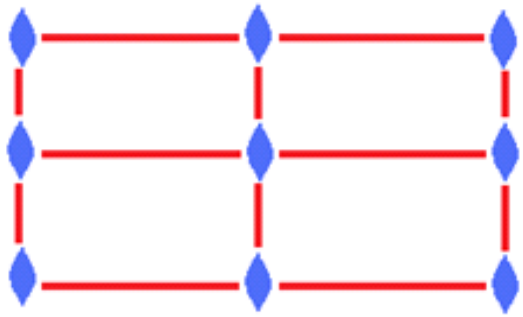




body-centered orthorhombic

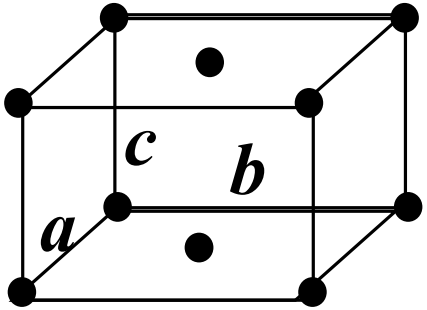
≠ based centered orthorhombic

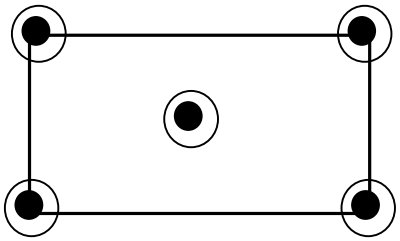
(2) Derived from centered rectangular lattice



$(a \neq b; \gamma = 90^{o})$

(a) C-face centered Orthorhombic

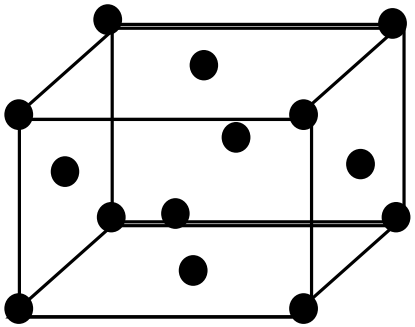


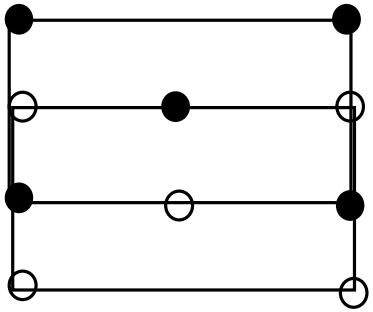


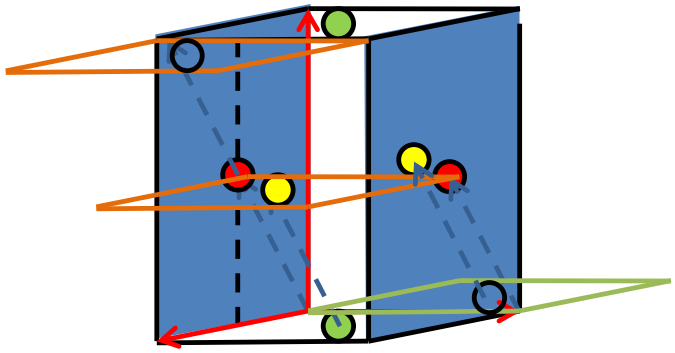
C- face centered orthorhombic

= B- face centered orthorhombic

(b) Face-centered Orthorhombic (F-cell)





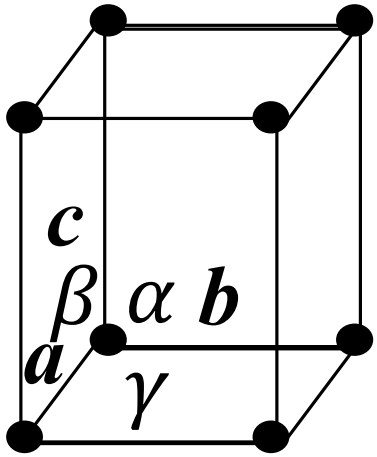


- Up & Down
- Left & Right
- Front & Back

Orthorhombic has 4 types

1. Primitive orthorhombic
2. Base centered orthorhombic
3. Body centered orthorhombic
4. Face centered orthorhombic

(4) Tetragonal system



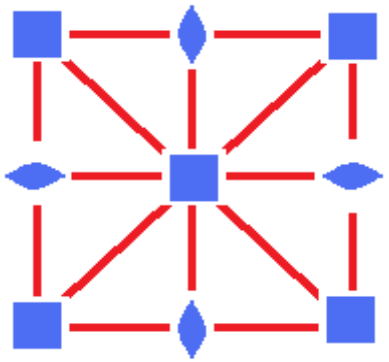
$(a = b \neq c;$
$\alpha = \beta = \gamma = 90^{o})$

One tetrad axis

starting from square lattice

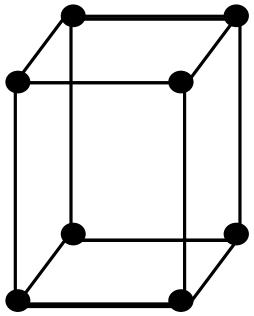
$(a = b; \gamma = 90^{o})$

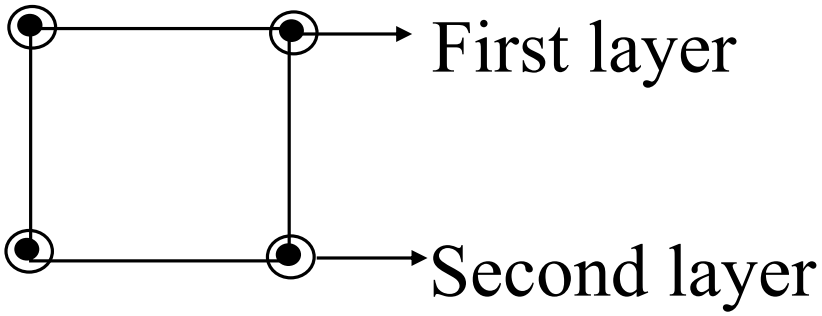
starting from square lattice $(a = b; \gamma = 90^{o})$



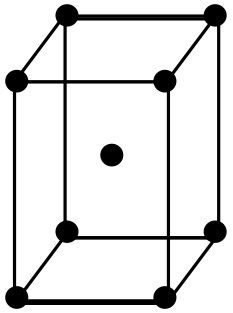
(1) maintain 4-fold symmetry

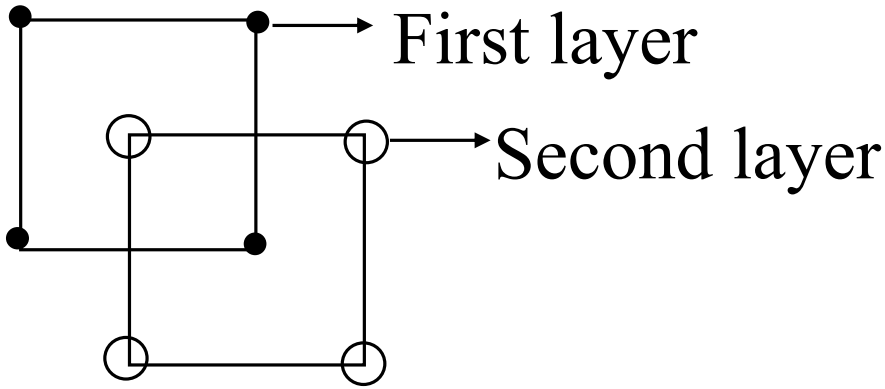
(a) Primitive tetragonal lattice





(b) Body-centered tetragonal lattice

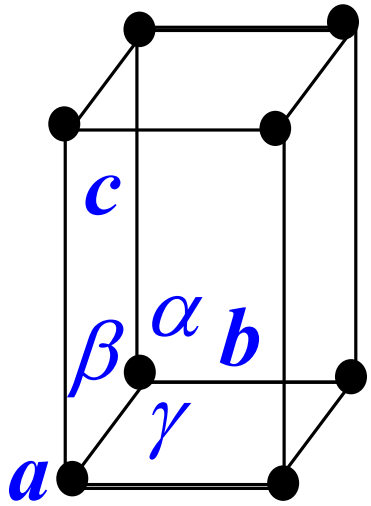




Tetragonal has 2 types

1. Primitive tetragonal
2. Body centered tetragonal

(5) Hexagonal system

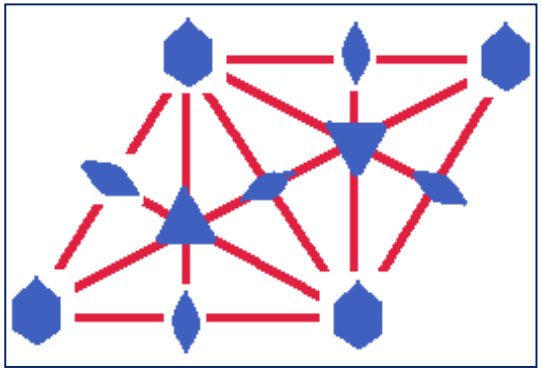


$$a = b \neq c;$$
$$\alpha = \beta = 90^{o};\ \gamma = 120^{o}$$

One hexad axis

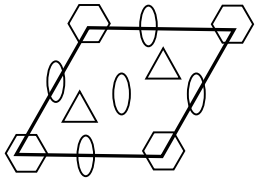
starting from hexagonal lattice (2D)

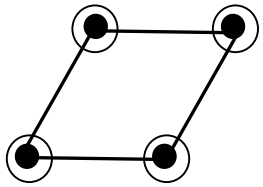
$$a = b;\ \gamma = 120^{o}$$

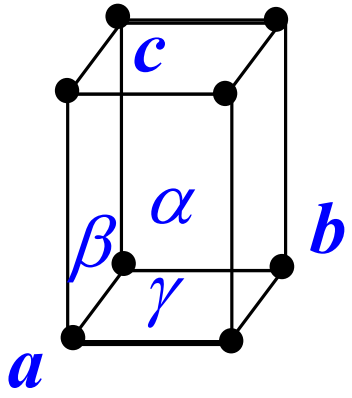


(1) maintain 6-fold symmetry

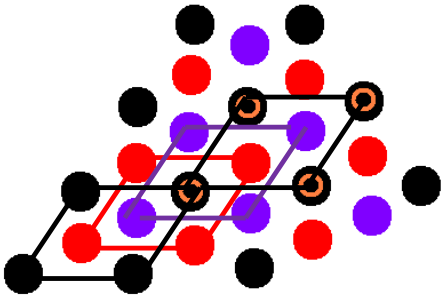
Primitive hexagonal lattice

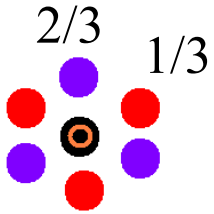


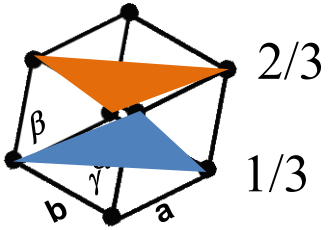


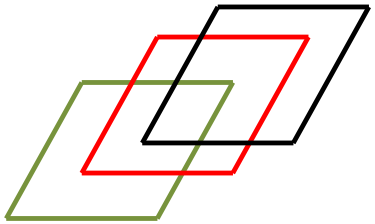


(2) maintain 3-fold symmetry









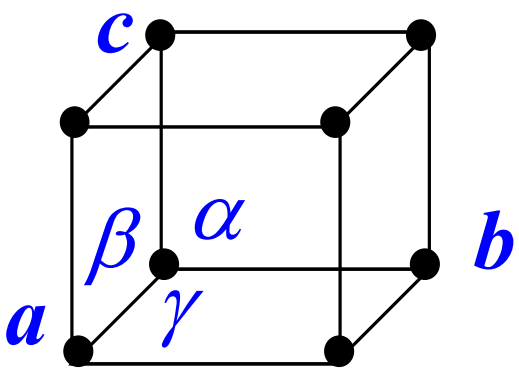
$a = b = c;$

$\alpha = \beta = \gamma \neq 90^{o}$

Hexagonal has 1 types
1. Primitive hexagonal

Rhombohedral (trigonal)
2. Primitive rhombohedral (trigonal)

(6) Cubic system



$a = b = c;$

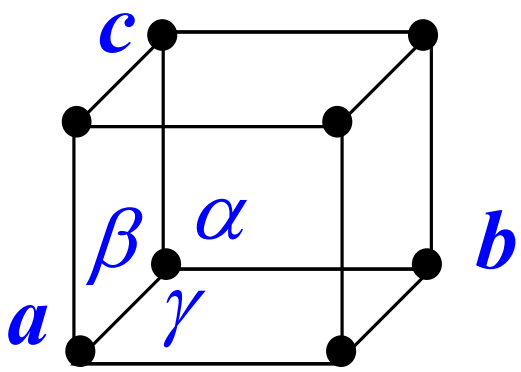
$\alpha = \beta = \gamma = 90^{o}$

4 triad axes (triad axis = cube diagonal)

Cubic is a special form of Rhombohedral lattice

Cubic system has 4 triad axes mutually inclined along cube diagonal

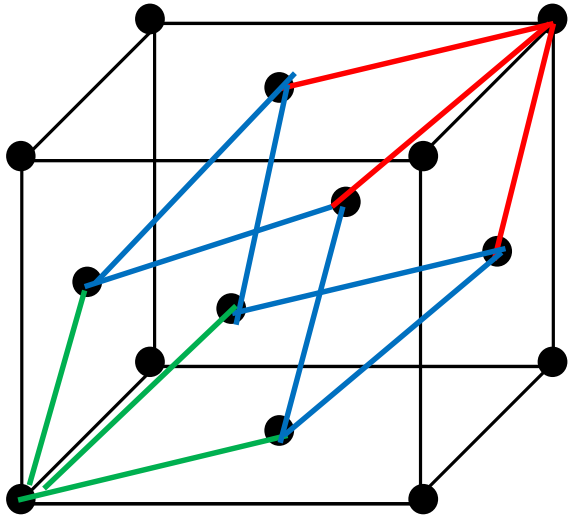
(a) Primitive cubic $\alpha = 90^{o}$



$\boldsymbol{a = b = c}$;

$\alpha = \beta = \gamma = 90^{o}$

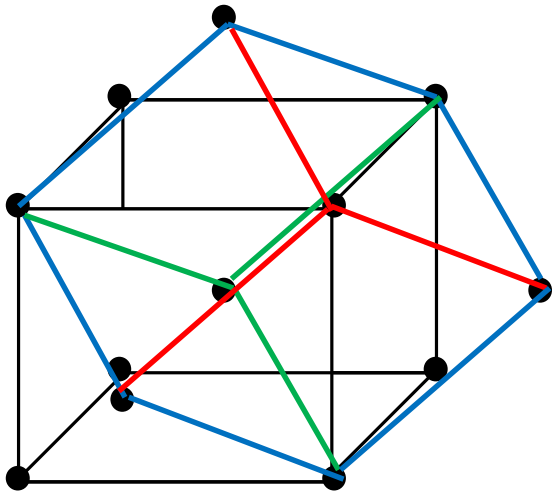
(b) Face centered cubic $\alpha = 60^{o}$



$\boldsymbol{a = b = c}$;

$\alpha = \beta = \gamma = 60^{o}$

(c) body centered cubic $\alpha = 109^{o}$



$\boldsymbol{a = b = c}$;

$\alpha = \beta = \gamma = 109^{o}$

cubic (isometric)

Special case of orthorhombic with $a = b = c$

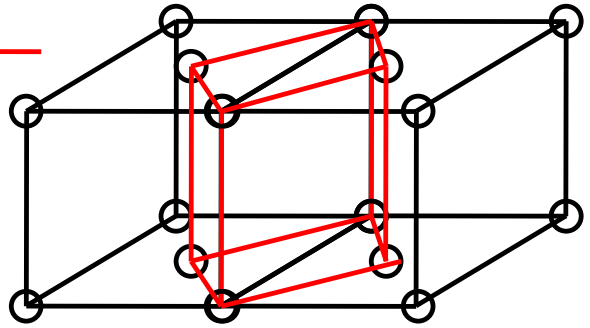
Primitive (P)

Body centered (I)

Face centered (F)

Base center (C)

Tetragonal (I)?



3 fold rotation disappears

Tetragonal (P) $a = b \neq c$

Cubic has 3 types

1. Primitive cubic (simple cubic)
2. Body centered cubic (BCC)
3. Face centered cubic (FCC)

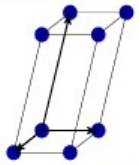
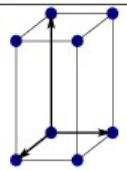
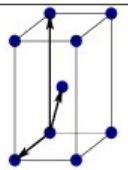
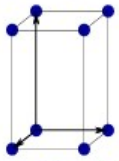
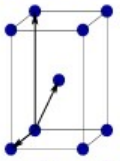
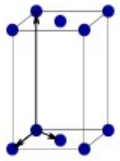
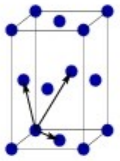
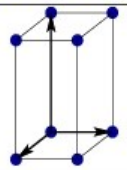
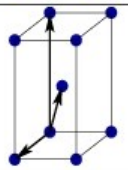
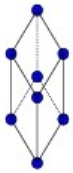
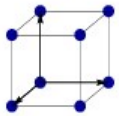
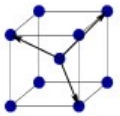
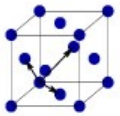
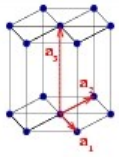
Bravais lattice	Parameters	Simple (P)	Volume centered (I)	Base centered (C)	Face centered (F)
Triclinic	$a_1 \neq a_2 \neq a_3$ $\alpha_{12} \neq \alpha_{23} \neq \alpha_{31}$				
Monoclinic	$a_1 \neq a_2 \neq a_3$ $\alpha_{23} = \alpha_{31} = 90°$ $\alpha_{12} \neq 90°$			√	
Orthorhombic	$a_1 \neq a_2 \neq a_3$ $\alpha_{12} = \alpha_{23} = \alpha_{31} = 90°$				
Tetragonal	$a_1 = a_2 \neq a_3$ $\alpha_{12} = \alpha_{23} = \alpha_{31} = 90°$			$= P$	$= I$
Trigonal	$a_1 = a_2 = a_3$ $\alpha_{12} = \alpha_{23} = \alpha_{31} < 120°$				
Cubic	$a_1 = a_2 = a_3$ $\alpha_{12} = \alpha_{23} = \alpha_{31} = 90°$			= T P	
Hexagonal	$a_1 = a_2 \neq a_3$ $\alpha_{12} = 120°$ $\alpha_{23} = \alpha_{31} = 90°$				

Table 1.1: Bravais lattices in three-dimensions.

http://www.theory.nipne.ro/~dragos/Solid/Bravais_table.jpg

III Crystal Symmetry

3-3 Point group and space group

A. Point group

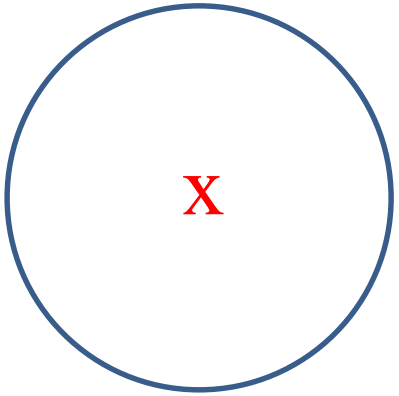
Symbols of the 32 three dimensional point groups

Next page!

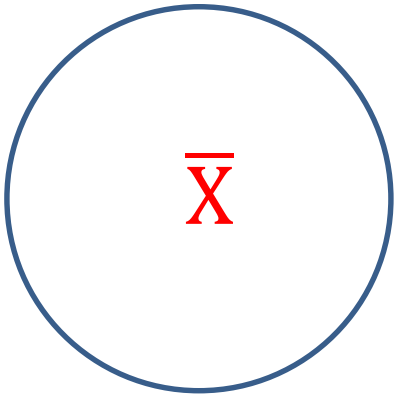
1, 2, 3, 4, 6, 23 + diad axis, mirror, inversion
$\rightarrow$ 32 point groups

General symbol	Triclinic	Monoclinic 1st setting	Tetragonal	Trigonal	Hexagonal	Cubic
X	1	2	4	3	6	23
$\bar{X}$		$\bar{2} \equiv m$	$\bar{4}$		$\bar{6}$	
X + centre (include $\bar{X}$ odd order)	$\bar{1}$	$\frac{2}{m}$ (2/*m*)	$\frac{4}{m}$	$\bar{3}$	$\frac{6}{m}$	$\frac{2}{m}\bar{3} \equiv m3$
	Monoclinic 2nd setting	Orthorho-mbic				
X2	$2 \equiv 12$	222	422	32	622	432
Xm	$m \equiv 1m$	*mm*2 (2*mm*)	4*mm*	3*m*	6*mm*	$\bar{4}3m$
$\bar{X}$2 or $\bar{X}$m even			$\bar{4}2m$		$\bar{6}m2$	
X2 + centre Xm + centre Include $\bar{X}$m odd order	$\frac{2}{m}$	$mmm \equiv \frac{2}{m}\frac{2}{m}\frac{2}{m}$	$\frac{4}{m}mm \equiv \frac{4}{m}\frac{2}{m}\frac{2}{m}$	$\bar{3}m \equiv \bar{3}\frac{2}{m}$	$\frac{6}{m}mm \equiv \frac{6}{m}\frac{2}{m}\frac{2}{m}$	$m3m \equiv \frac{4}{m}\bar{3}\frac{2}{m}$

1. Rotation axis X



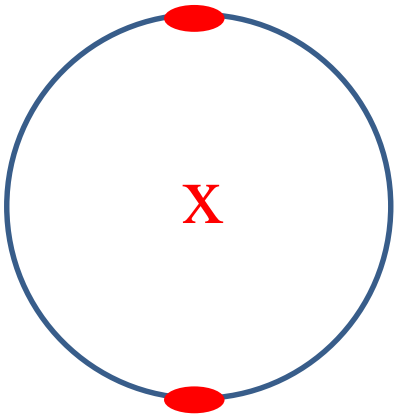
2. Rotation-Inversion axis $\bar{X}$

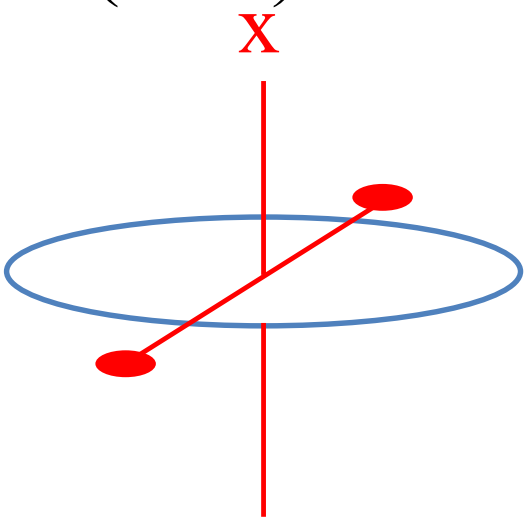


3. X + centre (inversion): Include $\bar{X}$ for odd order

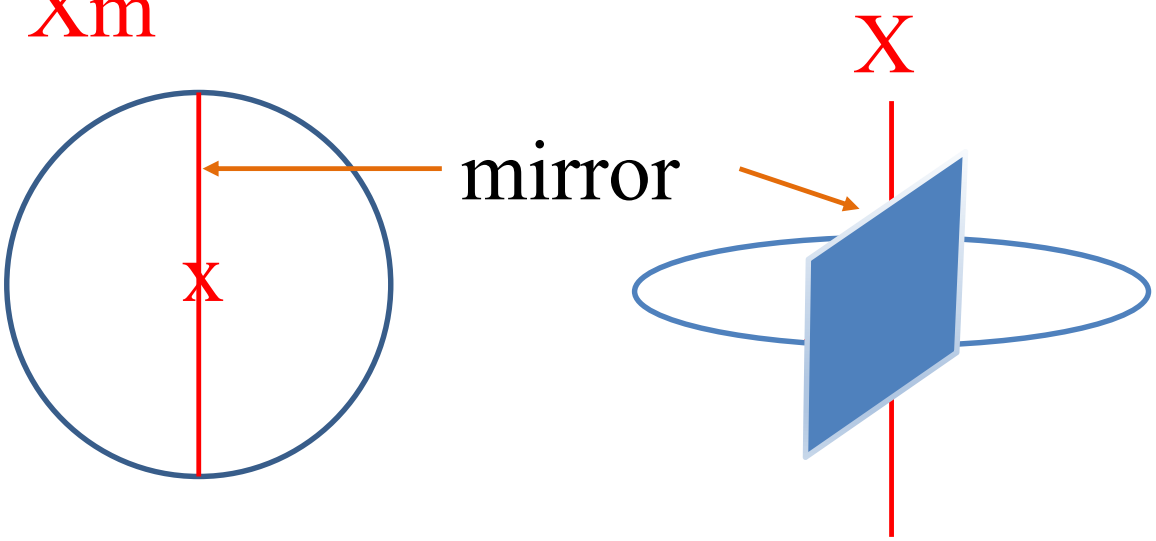
4. Rotation axis + diad axis (axes) normal to it:

X2





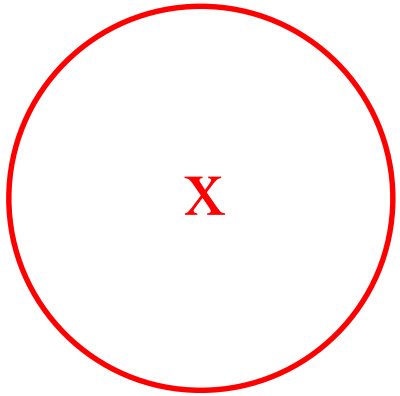
5. Rotation axis + mirror plane (planes) parallel to it: Xm

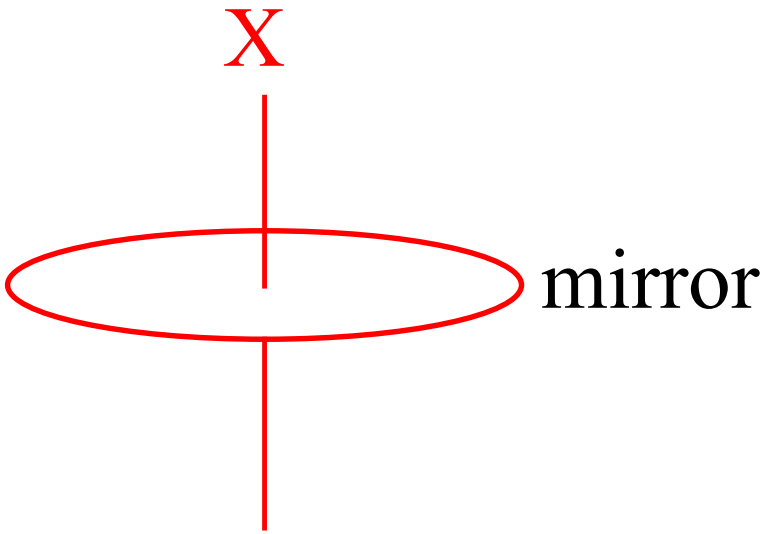


6. $\overline{X}2$ or $\overline{X}m$ even: only for even rotation symmetry (refer to 4 and 5 except the rotation axis is $\overline{X}$)

7. X2 + centre; Xm +centre: the same result (Include $\overline{X}m$ for odd order)

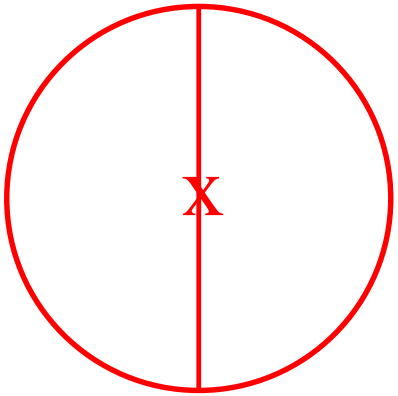
Rotation axis with mirror plane normal to it: X/m

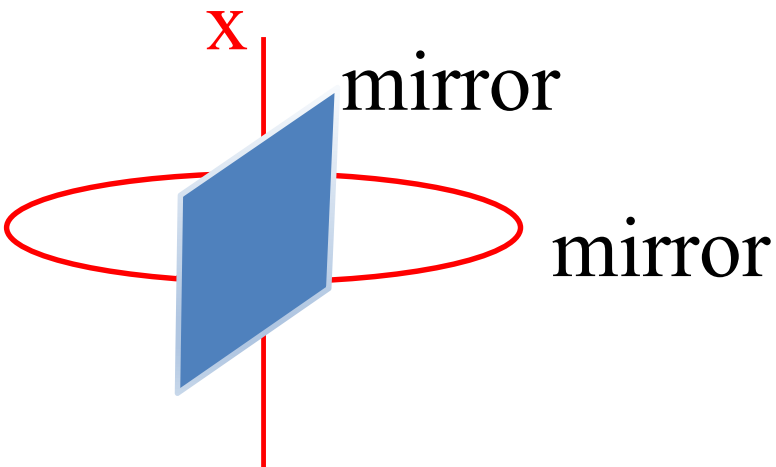


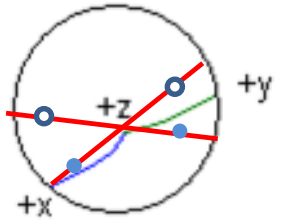
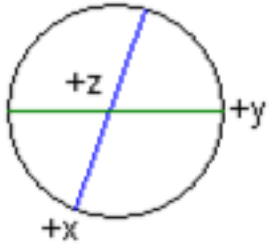
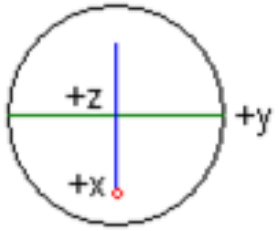


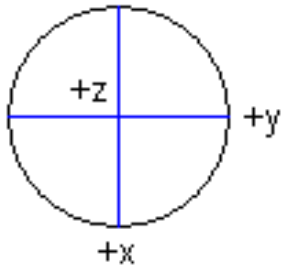
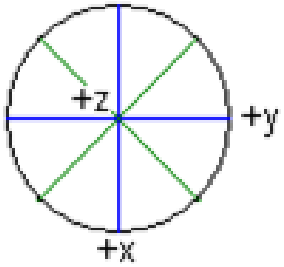
Rotation axis with mirror plane (planes) normal to it and mirror plane (planes) parallel to it X/mm

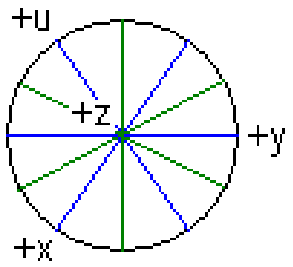
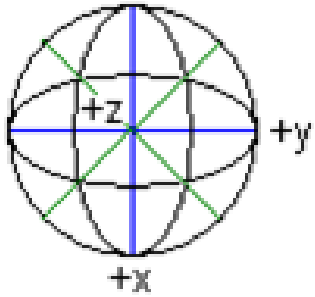
X/m





System and point group	Position in point group symbol			Stereographic representation
	Primary	Secondary	Tertiary	
Triclinic 1, $\bar{1}$	Only one symbol which denotes all directions in the crystal.			
Monoclinic 2, m, 2/m	The symbol gives the nature of the unique diad axis (rotation and/or inversion). 1st setting: z-axis unique 2nd setting: y-axis unique			 1st setting  2nd setting

System and point group	Position in point group symbol			Stereographic representation
	Primary	Secondary	Tertiary	
Orthorhombic 222, mm2, mmm	Diad (rotation and/or inversion) along x-axis	Diad (rotation and/or inversion) along y-axis	Diad (rotation and/or inversion) along z-axis	
Tetragonal 4, $\bar{4}$, 4/m, 422, 4mm, $\bar{4}$2m, 4/mmm	Tetrad (rotation and/or inversion) along z-axis	Diad (rotation and/or inversion) along x- and y-axes	Diad (rotation and/or inversion) along [110] and [1$\bar{1}$0] axis	

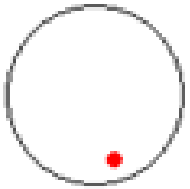
System and point group	Position in point group symbol			Stereographic representation
	Primary	Secondary	Tertiary	
Trigonal and Hexagonal 3, $\bar{3}$, 32, 3m, $\bar{3}$m, 6, $\bar{6}$, 6/m, 622, 6mm, $\bar{6}$m2, 6/mmm	Triad or hexad (rotation and/or inversion) along z-axis	Diad (rotation and/or inversion) along x-, y- and u-axes	Diad (rotation and/or inversion) normal to x-, y-, u-axes in the plane (0001)	
Cubic 23, m3, 432,$\bar{4}$3m, m3m	Diads or tetrad (rotation and/or inversion) along <100> axes	Triads (rotation and/or inversion) along <111> axes	Diads (rotation and/or inversion) along <110> axes	

Crystal System	Symmetry Direction		
	Primary	Secondary	Tertiary
Triclinic	None		
Monoclinic	[010]		
Orthorhombic	[100]	[010]	[001]
Tetragonal	[001]	[100]/[010]	[110]
Hexagonal	[001]	[100]	[120]
Trigonal	[111]	[010]	$[1\bar{1}0]$
Cubic	[100]/[010]/ [001]	[111]	[110]

Triclinic

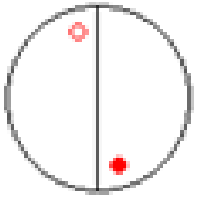
Rotation axis X

1



Rotation-Inversion axis $\bar{X}$

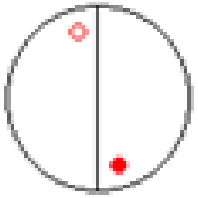
$\bar{1}$



X + centre

Include $\bar{X}$ (odd order)

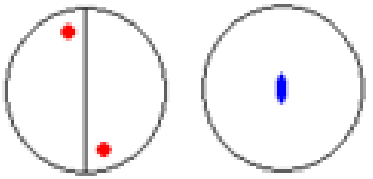
$\bar{1}$



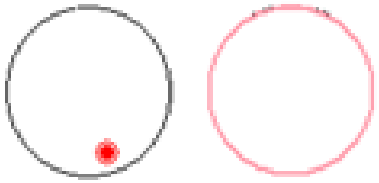
For odd order includes $\bar{X}$ already!

Monoclinic
1st setting

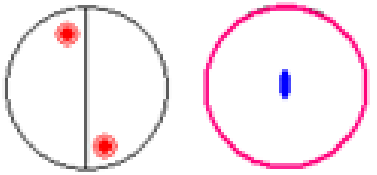
X — 2

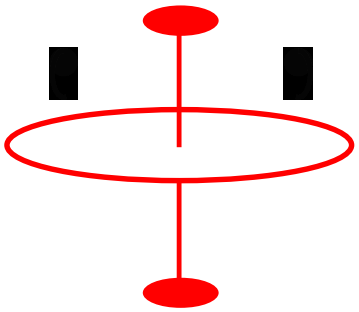


$\bar{X}$ — $\bar{2} = m$



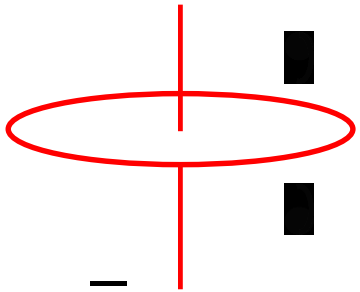
X + centre
Include $\bar{X}$ (odd order) — $\frac{2}{m}$





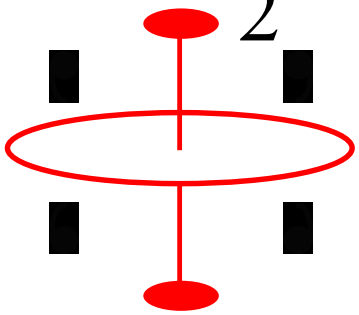
2

mirror



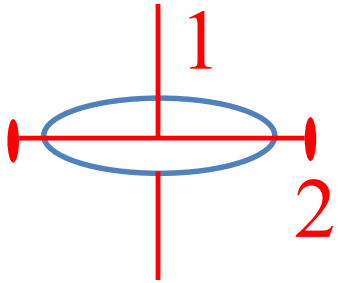
$\bar{2} = m$

mirror



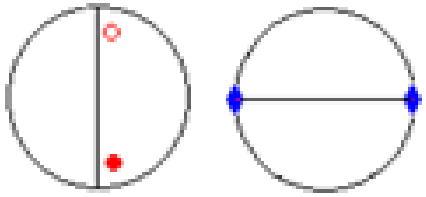
$\frac{2}{m}$

Monoclinic
2st setting



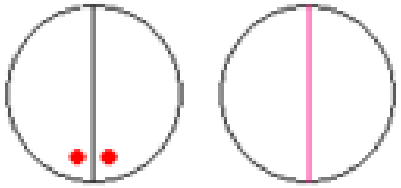
X2

12 ≡ 2



Xm

1m ≡ m

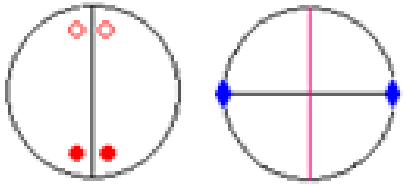


$\bar{X}2$ or $\bar{X}m$ (even)

2/m

X2 + centre, Xm +centre
Include $\bar{X}m$ (odd order)

2/m



Orthorhombic

Rotation axis X

Rotation-Inversion axis $\bar{X}$

X + centre

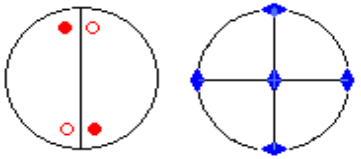
Include $\bar{X}$ (odd order)

2

Included in monoclinic system

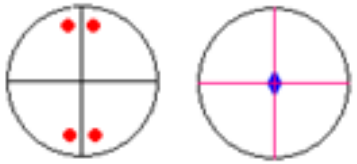
X2

222



Xm

2*mm* (2D) = *mm2*



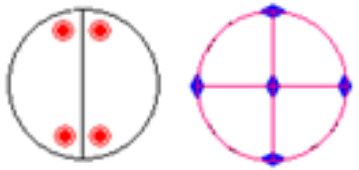
$\bar{X}2$ or $\bar{X}m$ even

mm2

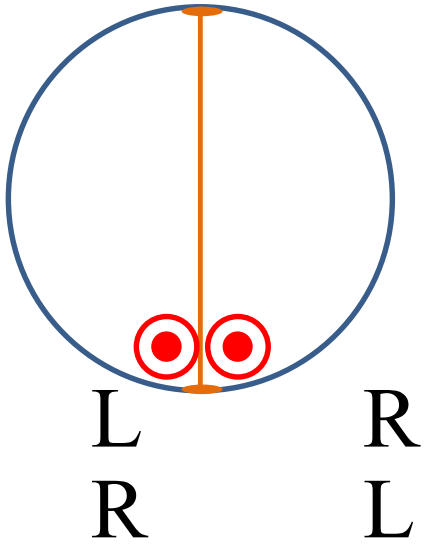
X2 + centre, Xm +centre

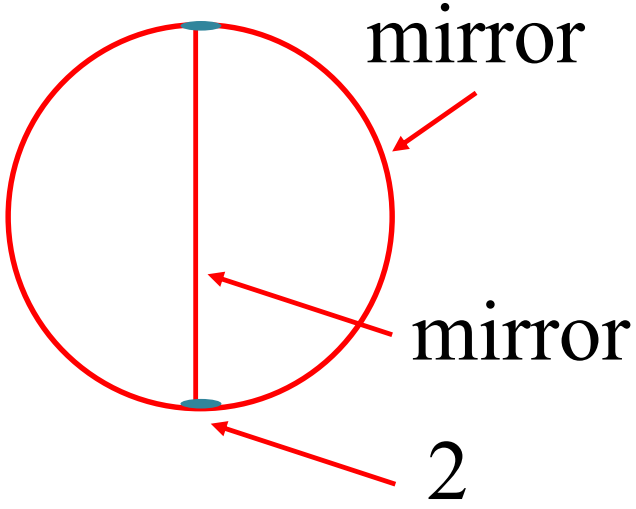
Include $\bar{X}m$ (odd order)

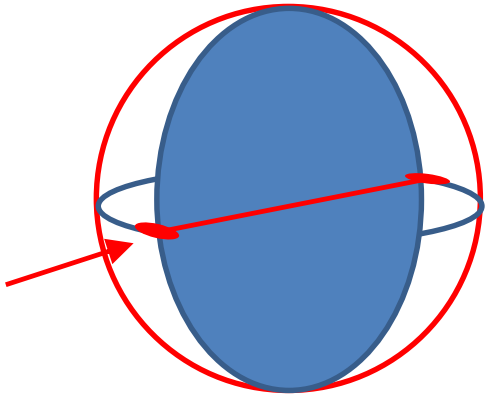
mmm ≡ 2/*m*2/*m*2/*m*

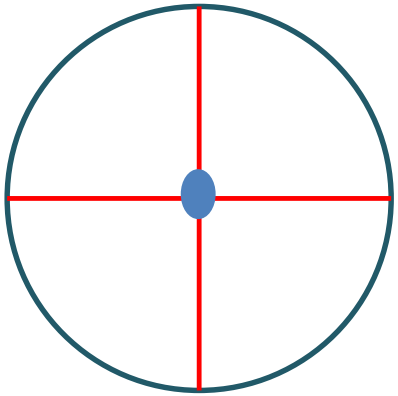


$\bar{2}2$ or $\bar{2}m$









2mm

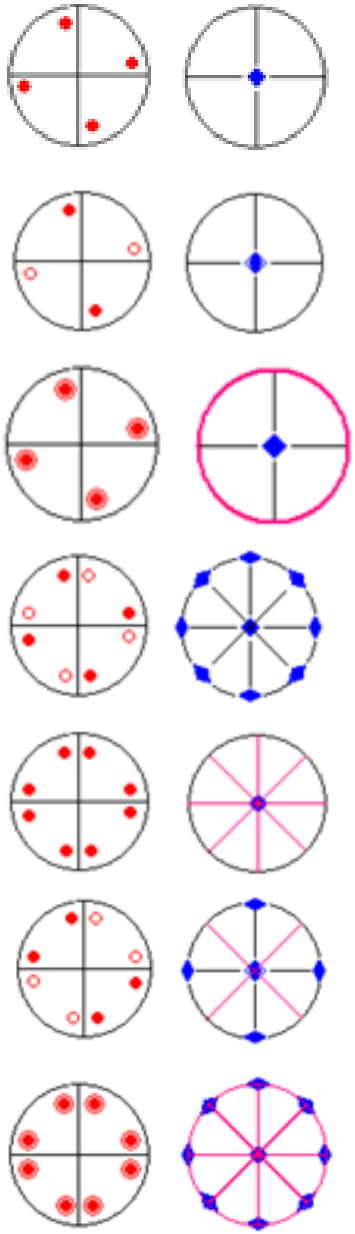
Question: $mmm \equiv 2/m2/m2/m$

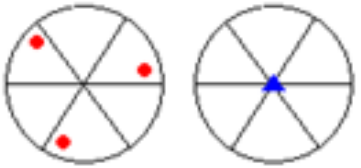
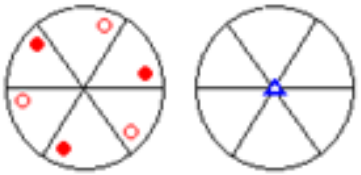
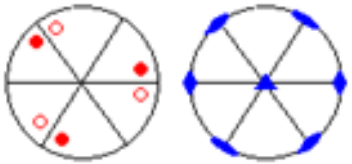
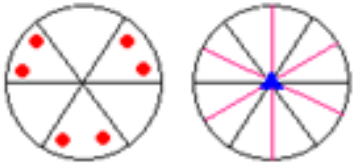
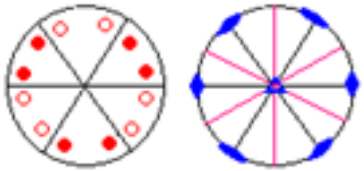
new old terminology

Using *mmm* symmetry operation, one can get the same point group as $\frac{2}{m}\frac{2}{m}\frac{2}{m}$!

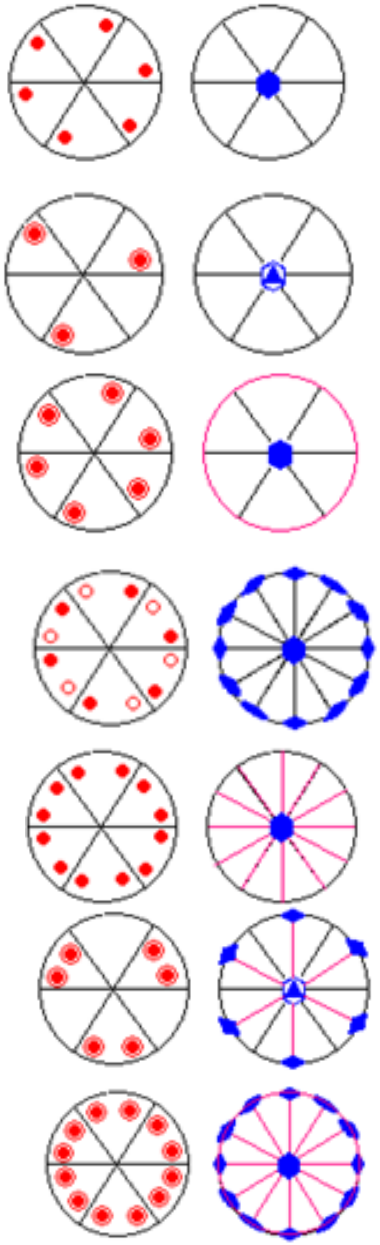
Tetragonal

Rotation axis X	4
Rotation-Inversion axis $\bar{X}$	$\bar{4}$
X + centre Include $\bar{X}$ (odd order)	$\frac{4}{m}$
X2	422
Xm	$4mm$
$\bar{X}2$ or $\bar{X}m$ even	$\bar{4}2m$
X2 + centre, Xm +centre Include $\bar{X}m$ (odd order)	$4/mmm \equiv$ $4/m\ 2/m\ 2/m$



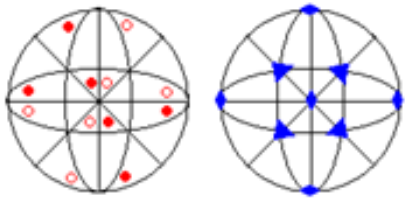
	Trigonal	
Rotation axis X	3	
Rotation-Inversion axis $\bar{X}$	$\bar{3}$	
X + centre Include $\bar{X}$ (odd order)	$\bar{3}$	
X2	32	
Xm	$3m$	
$\bar{X}$2 or $\bar{X}$m even	↓	
X2 + centre, Xm +centre Include $\bar{X}$m (odd order)	$\bar{3}m \equiv$ $\bar{3}\,2/m$	

	Hexagonal
Rotation axis X	6
Rotation-Inversion axis $\overline{X}$	$\bar{6}$
X + centre Include $\overline{X}$ (odd order)	$\frac{6}{m}$
X2	622
Xm	$6mm$
$\overline{X}2$ or $\overline{X}m$ even	$\bar{6}m2$
X2 + centre, Xm +centre Include $\overline{X}m$ (odd order)	6/mmm ≡ 6/m 2/m 2/m



Cubic

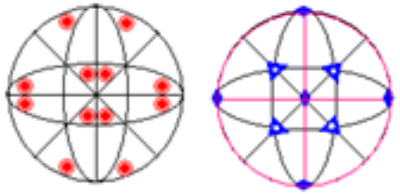
Rotation axis X — 23



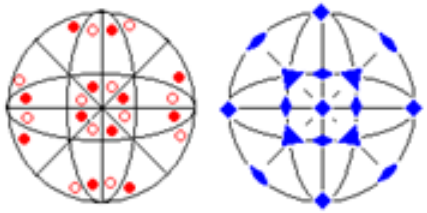
Rotation-Inversion axis $\bar{X}$ —



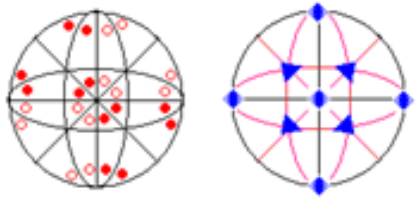
X + centre
Include $\bar{X}$ (odd order) — $m3 \equiv 2/m\ \bar{3}$



X2 (+2 [110] direction) — 432



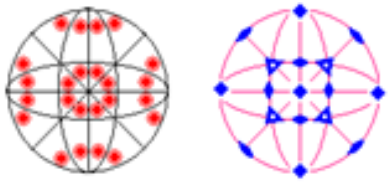
Xm (+m [110] direction) — $\bar{4}3m$



$\bar{X}2$ or $\bar{X}m$ even —



X2 + centre, Xm +centre
Include $\bar{X}m$ (odd order) — m3m ≡ 4/m $\bar{3}$ 2/m



B. Space group

14 lattices + 32 point groups
→ 230 Space groups

Crystal Class	Bravais Lattices	Point Groups
Triclinic	P	1, $\bar{1}$
Monoclinic	P, C	2, m, 2/m
Orthorhombic	P, C, F, I	222, mm2, 2/m 2/m 2/m
Trigonal	P, R	3, $\bar{3}$, 32, 3m, $\bar{3}$2/m
Hexagonal	P	6, $\bar{6}$, 6/m, 622, 6mm, $\bar{6}$m2, 6/m 2/m 2/m
Tetragonal	P, I	4, $\bar{4}$, 4/m, 422, 4mm, $\bar{4}$2m, 4/m 2/m 2/m
Isometric	P, F, I	23, 2/m$\bar{3}$, 432, $\bar{4}$3m, 4/m$\bar{3}$2/m

Table for all space groups
Look at the notes!

Good web site to read about space group

http://www.uwgb.edu/dutchs/SYMMETRY/3dSpaceGrps/3dspgrp.htm

http://img.chem.ucl.ac.uk/sgp/mainmenu.htm

Symmetry elements in space group

(1)Point group

(2)Translation symmetry + point group

Translational symmetry operations

The first character:

P: primitive

A, B, C: A, B, C-base centered

F: Face centered

I: Body centered

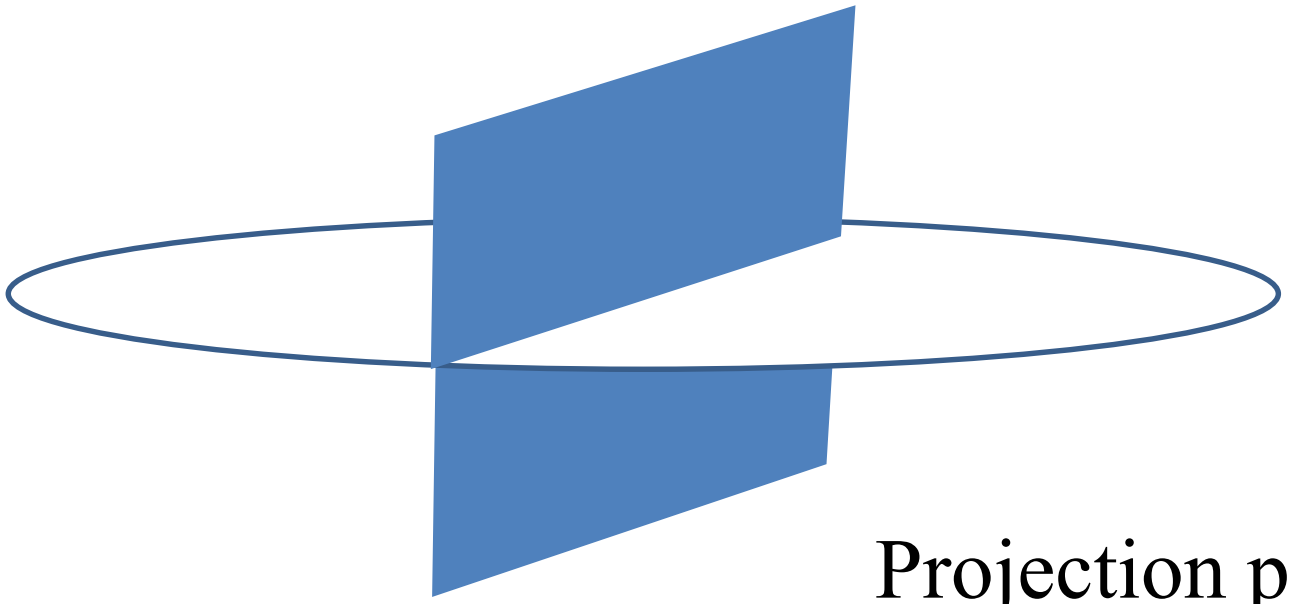
R: Romohedral

Glide plane also exists for 3D space group with more possibility

Symmetry planes normal to the plane of projection

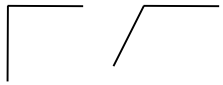
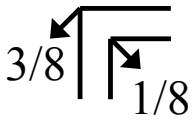
Symmetry plane	Graphical symbol	Translation	Symbol
Reflection plane		None	*m*
Glide plane		1/2 along line	*a*, *b*, or *c*
Glide plane		1/2 normal to plane	*a*, *b*, or *c*
Double glide plane		1/2 along line & 1/2 normal to plane	*e*
Diagonal glide plane		1/2 along line & 1/2 normal to plane	*n*
Diamond glide plane		1/4 along line & 1/4 normal to plane	*d*

Symmetry planes normal to the plane of projection



Projection plane

Symmetry planes parallel to plane of projection

Symmetry plane	Graphical symbol	Translation	Symbol
Reflection plane		None	*m*
Glide plane		1/2 along arrow	*a*, *b*, or *c*
Double glide plane		1/2 along either arrow	*e*
Diagonal glide plane		1/2 along the arrow	*n*
Diamond glide plane		1/8 or 3/8 along the arrows	*d*

The presence of a *d*-glide plane automatically implies a centered lattice!

Glide planes

---- translation plus reflection across the glide plane

* axial glide plane (glide plane along axis)

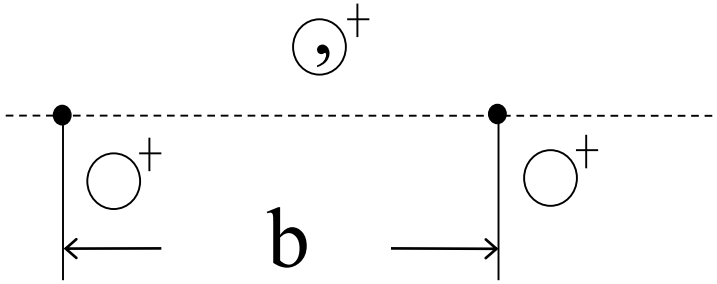
---- translation by half lattice repeat plus reflection

---- three types of axial glide plane

i. a glide, b glide, c glide (a, b, c)

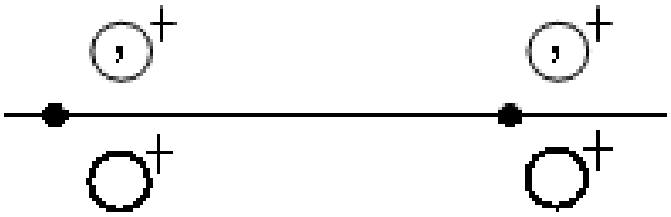
$\frac{1}{2}$ along line in plane $\equiv \frac{1}{2}$ along line parallel to projection plane

e.g. b glide



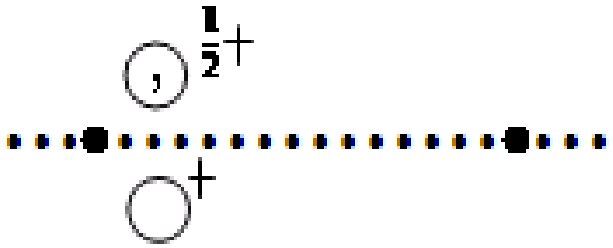
--- graphic symbol for the axial glide plane along y axis

c.f. mirror (m)

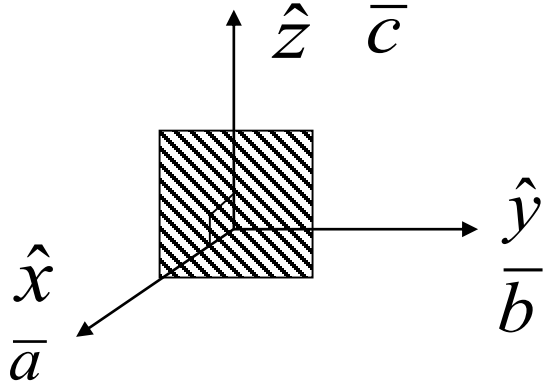


— graphic symbol for mirror

If the axial glide plane is $\frac{1}{2}$ normal to projection plane, the graphic symbol change to



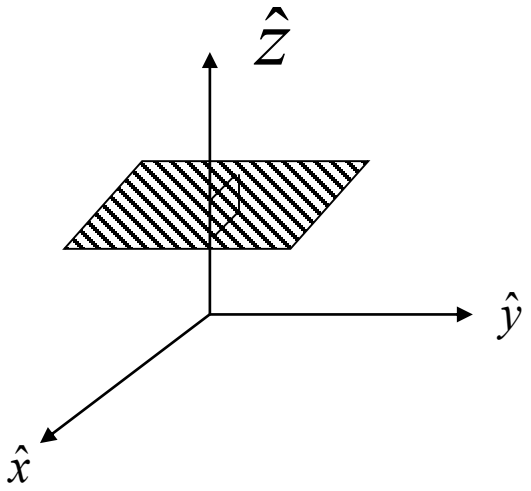
c glide

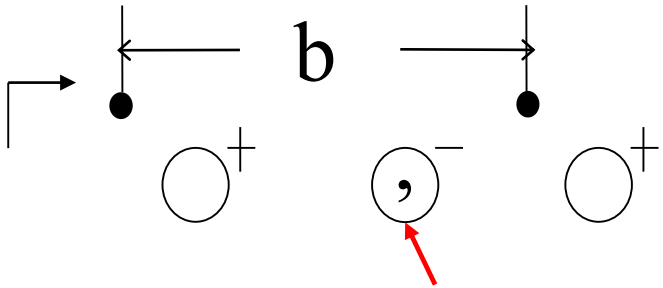


glide plane$\perp\hat{z}$ axis

If b glide plane is $\perp\hat{z}$ axis

glide plane symbol





c glide: $\frac{c}{2}$ along z axis

or

$\frac{a+b+c}{2}$ along [111] on rhombohedral axis

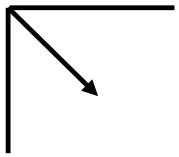
ii. Diagonal glide (n)

$\frac{a+b}{2}$, $\frac{b+c}{2}$, $\frac{a+c}{2}$ or $\frac{a+b+c}{2}$ (tetragonal, cubic system)

If glide plane is perpendicular to the drawing plane (*xy* plane), the graphic symbol is



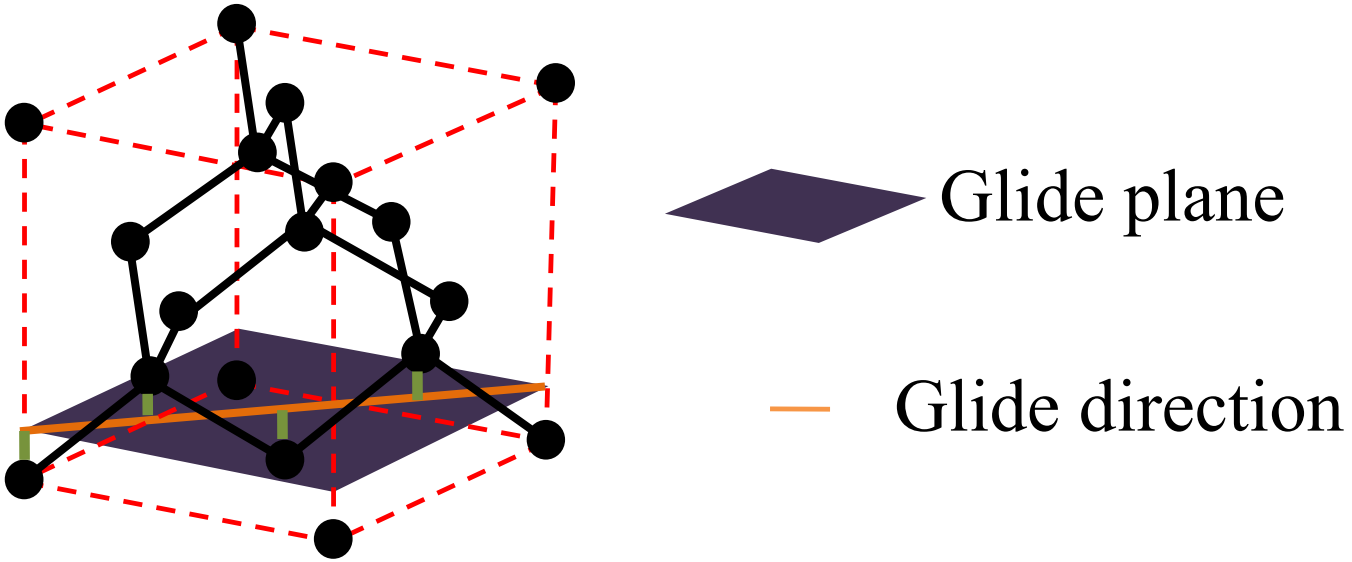
If glide plane is parallel to the drawing plane, the graphic symbol is



iii. Diamond glide (d)

$\frac{a+b}{4}, \frac{a+b+c}{4}$ (tetragonal, cubic system)





Symbols of symmetry axes

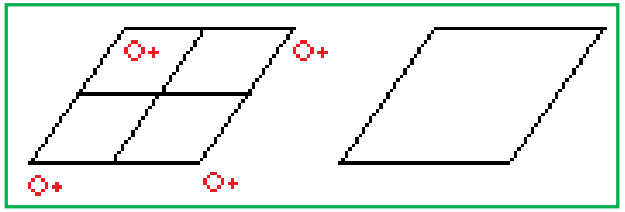
Symmetry Element	Graphical Symbol	Translation	Symbol
Identity	None	None	1
2-fold ⊥ page		None	2
2-fold in page		None	2
2 sub 1 ⊥ page		1/2	2_1
2 sub 1 in page		1/2	2_1
3-fold		None	3
3 sub 1		1/3	3_1
3 sub 2		2/3	3_2
4-fold		None	4
4 sub 1		1/4	4_1
4 sub 2		1/2	4_2
4 sub 3		3/4	4_3
6-fold		None	6
6 sub 1		1/6	6_1
6 sub 2		1/3	6_2
6 sub 3		1/2	6_3

Symmetry Element	Graphical Symbol	Translation	Symbol
6 sub 4		2/3	6_4
6 sub 5		5/6	6_5
Inversion		None	1
3 bar		None	3
4 bar		None	4
6 bar		None	$6 = 3/m$
2-fold and inversion		None	$2/m$
2 sub 1 and inversion		None	$2_1/m$
4-fold and inversion		None	$4/m$
4 sub 2 and inversion		None	$4_2/m$
6-fold and inversion		None	$6/m$
6 sub 3 and inversion		None	$6_3/m$

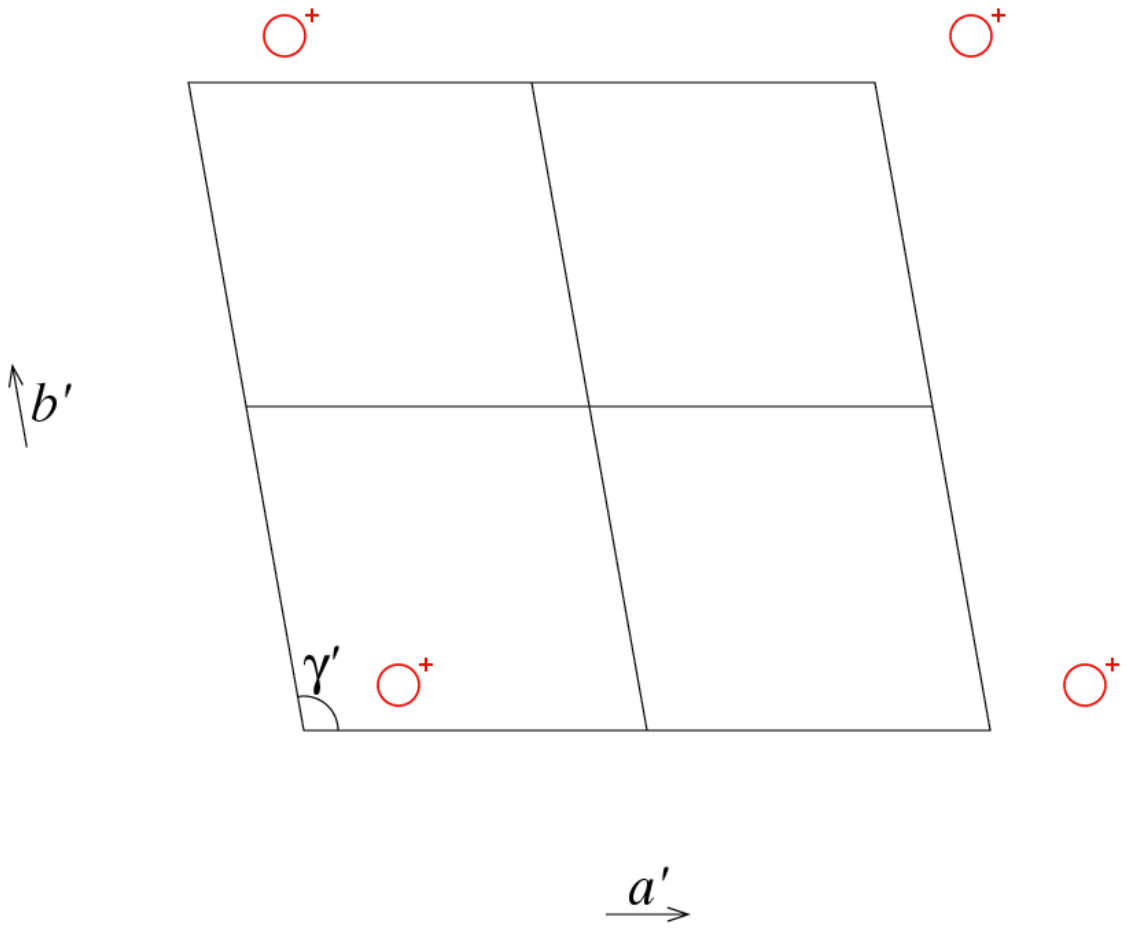
How to quickly identify the crystal class of a space group?

- Cubic – The secondary symmetry symbol will always be either 3 or –3 (i.e. Ia3, Pm3m, Fd3m)
- Tetragonal – The primary symmetry symbol will always be either 4, (-4), 4_1, 4_2 or 4_3 (i.e. $P4_12_12$, I4/m, P4/mcc)
- Hexagonal – The primary symmetry symbol will always be a 6, (-6), 6_1, 6_2, 6_3, 6_4 or 6_5 (i.e. P6mm, $P6_3$/mcm)
- Trigonal – The primary symmetry symbol will always be a 3, (-3) 3_1 or 3_2 (i.e P31m, R3, R3c, P312)
- Orthorhombic – All three symbols following the lattice descriptor will be either mirror planes, glide planes, 2-fold rotation or screw axes (i.e. Pnma, $Cmc2_1$, Pnc2)
- Monoclinic – The lattice descriptor will be followed by either a single mirror plane, glide plane, 2-fold rotation or screw axis or an axis/plane symbol (i.e. Cc, P2, $P2_1$/n)
- Triclinic – The lattice descriptor followed by either a 1 or a (-1).

Examples
Space group P1

P1 C_1^1		No. 1	P1	1 Triclinic
				
			Origin on 1	
Number of positions	Wyckoff notation	Point symmetry	Coordinates of equivalent positions	Condition limiting possible reflections
1	a	1	x, y, z	No conditions

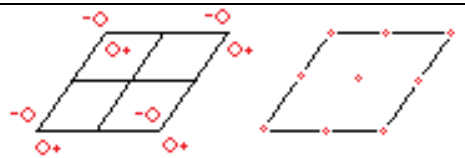
*P*1 *P* 1 1 No. 1



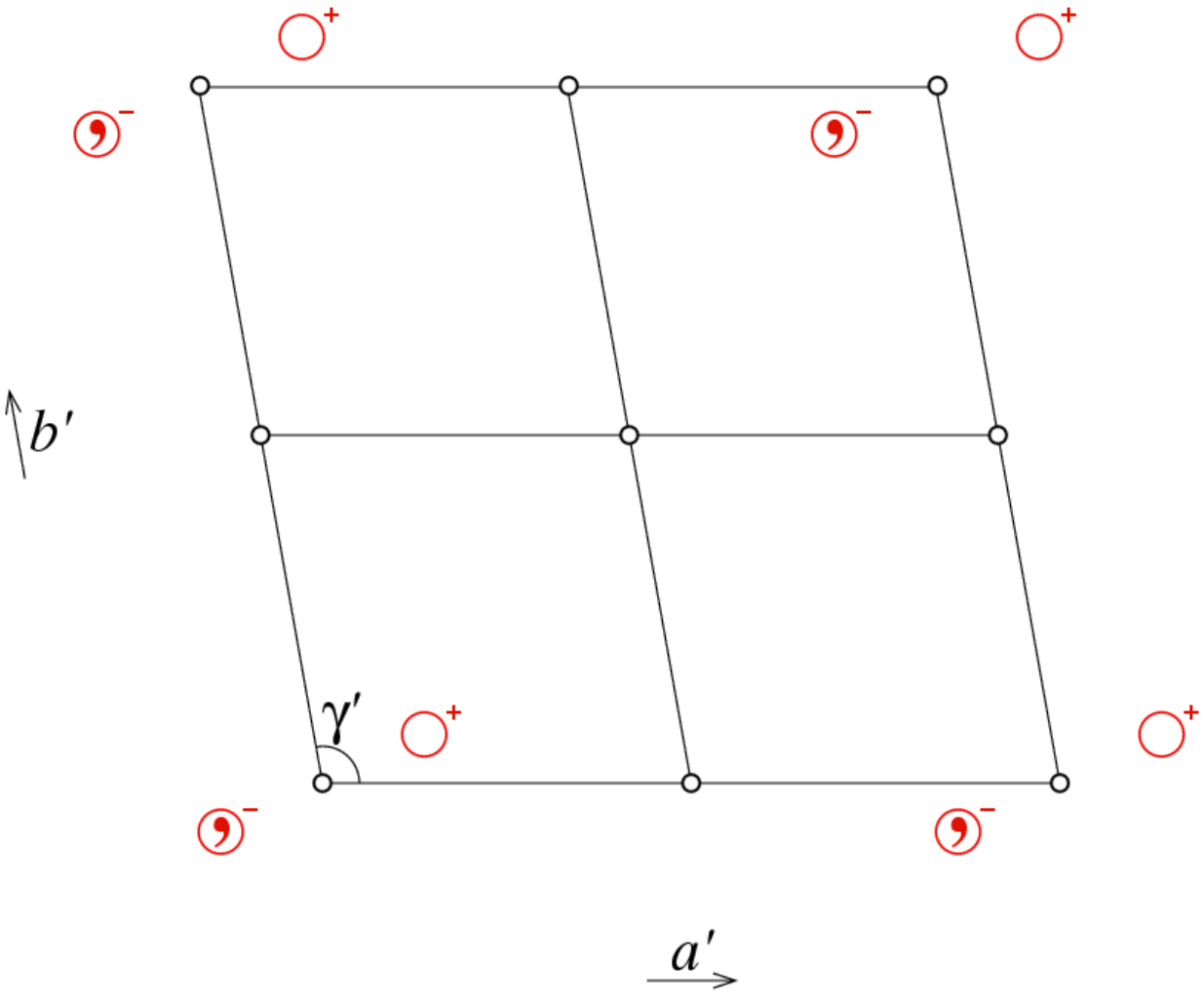
1 *x*, *y*, *z*

http://img.chem.ucl.ac.uk/sgp/large/001az1.htm

Space group $P\bar{1}$

$P\bar{1}$ C_i^1		No. 2	$P\bar{1}$	$\bar{1}$ Triclinic
				
			Origin on $\bar{1}$	
Number of positions	Wyckoff notation	Point symmetry	Coordinates of equivalent positions	Condition limiting possible reflections
2	i	1	x, y, z ; $\bar{x}$, $\bar{y}$, $\bar{z}$	General: No conditions
1	h	$\bar{1}$	$\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$	Special: No conditions
1	g	$\bar{1}$	$0, \frac{1}{2}, \frac{1}{2}$	
1	f	$\bar{1}$	$\frac{1}{2}, 0, \frac{1}{2}$	
1	e	$\bar{1}$	$\frac{1}{2}, \frac{1}{2}, 0$	
1	d	$\bar{1}$	$\frac{1}{2}, 0, 0$	
1	c	$\bar{1}$	$0, \frac{1}{2}, 0$	
1	b	$\bar{1}$	$0, 0, \frac{1}{2}$	
1	a	$\bar{1}$	0, 0, 0	

$P\bar{1}$ $P\ \bar{1}$ $\bar{1}$ No. 2

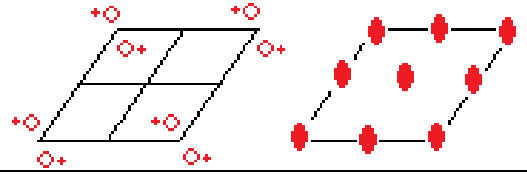


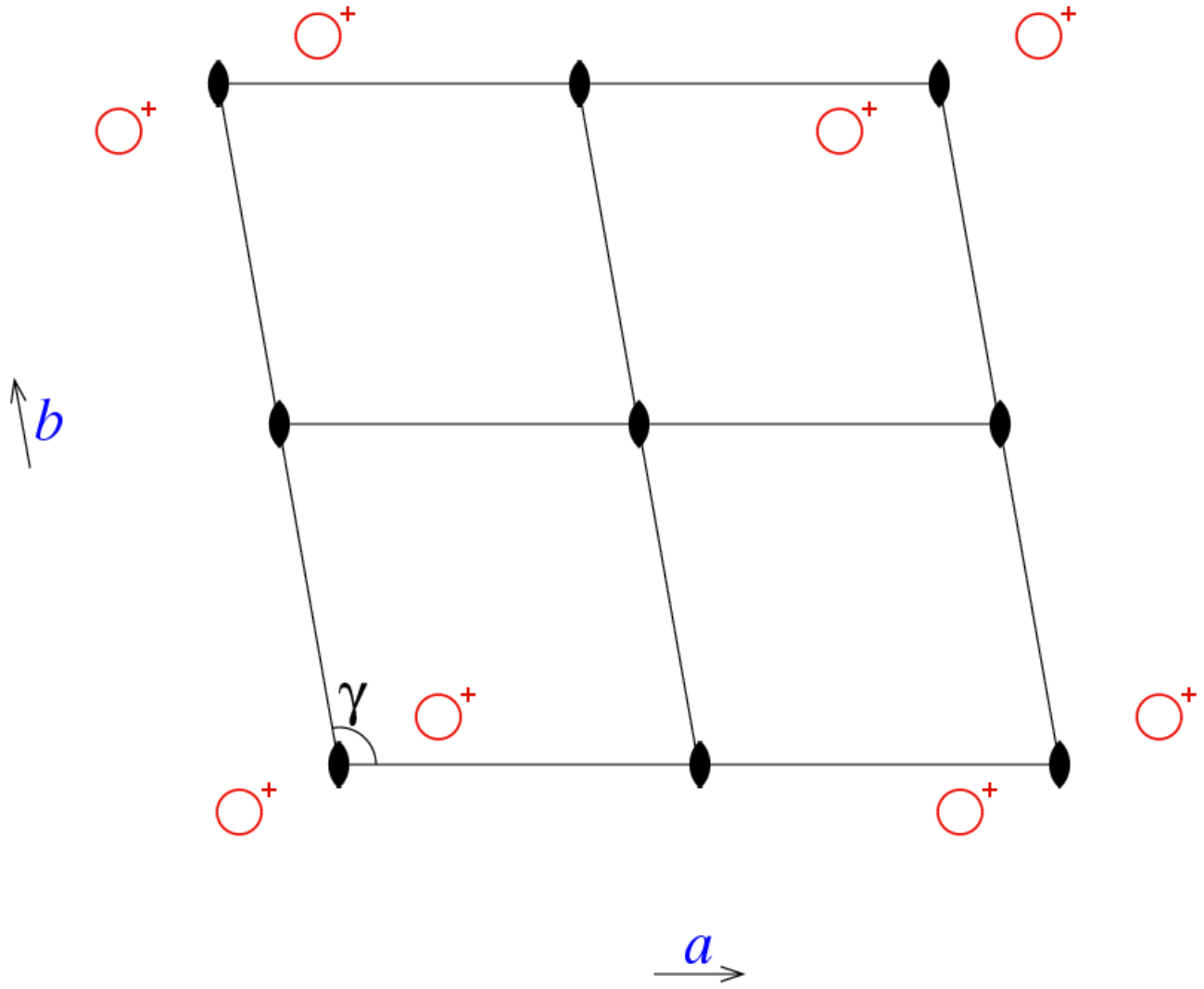
1 x, y, z

2 $\bar{x}, \bar{y}, \bar{z}$

http://img.chem.ucl.ac.uk/sgp/large/002az1.htm

Space group P112

$P112$ C_2^1		No. 3	P112	2 Monoclinic
				
Ist setting			Origin on 2; unique axis c	
Number of positions	Wyckoff notation	Point symmetry	Coordinates of equivalent positions	Condition limiting possible reflections
2	e	1	x, y, z; $\bar{x}$, $\bar{y}$, z	General: $\begin{Bmatrix} hkl \\ hk0 \\ 00l \end{Bmatrix}$ No conditions
1	d	2	$\frac{1}{2}$, $\frac{1}{2}$, z	Special: No conditions
1	c	2	$\frac{1}{2}$, 0, z	
1	b	2	0, $\frac{1}{2}$, z	
1	a	2	0, 0, z	

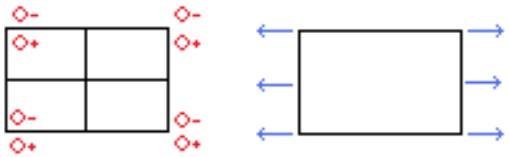


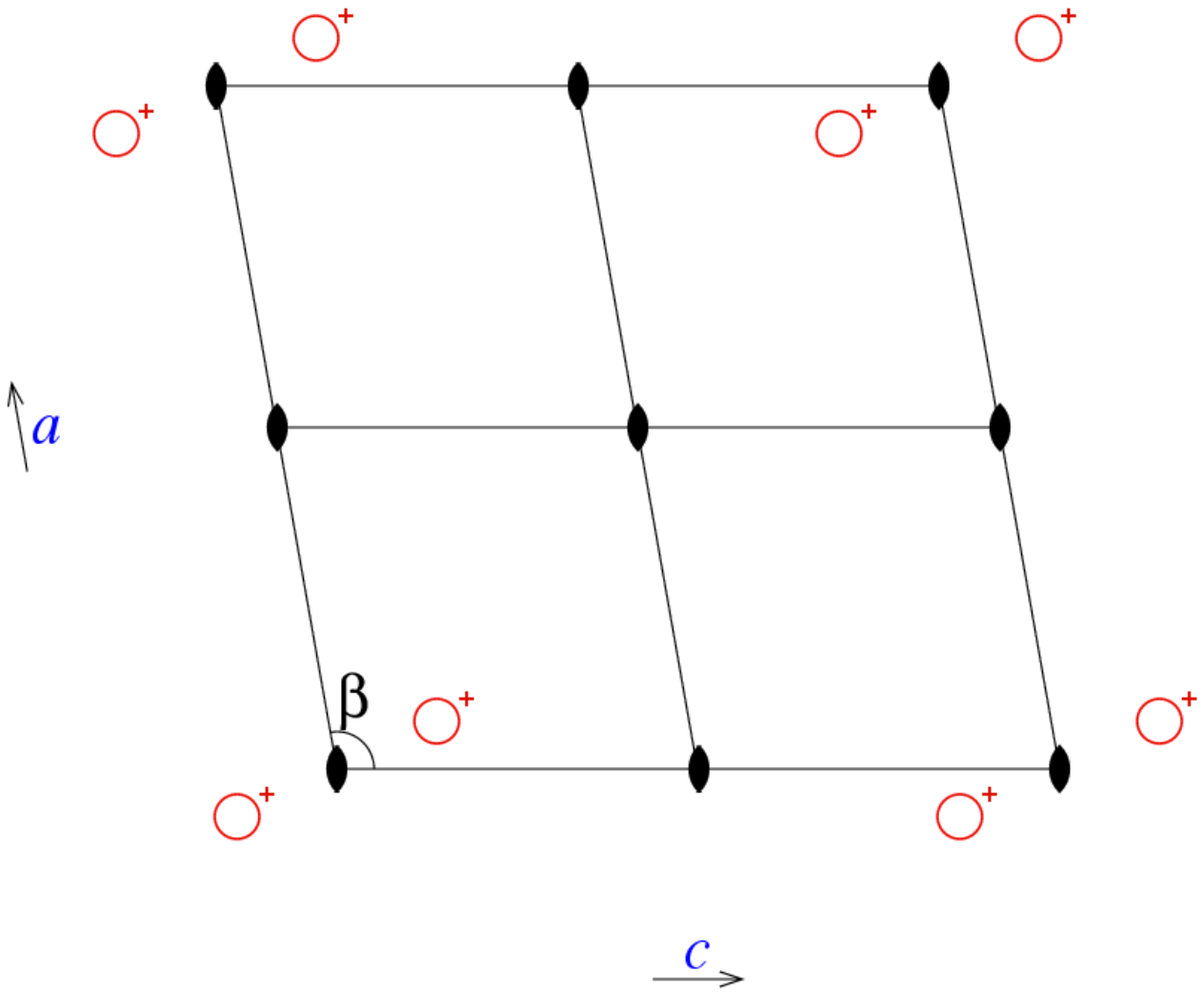
1 x, y, z

2 $\bar{x}, \bar{y}, z$

http://img.chem.ucl.ac.uk/sgp/large/003az1.htm

Space group P121

$P121$ C_2^1		No. 3	P121	2 Monoclinic
				
			Origin on 2; unique axis b	2nd setting
Number of positions	Wyckoff notation	Point symmetry	Coordinates of equivalent positions	Condition limiting possible reflections
2	e	1	x, y, z; $\bar{x}$, y, $\bar{z}$	General: $\begin{Bmatrix} hkl \\ h0l \\ 0k0 \end{Bmatrix}$ No conditions
1	d	2	$\frac{1}{2}$, y, $\frac{1}{2}$	Special: No conditions
1	c	2	$\frac{1}{2}$, y, 0	
1	b	2	0, y, $\frac{1}{2}$	
1	a	2	0, y, 0	

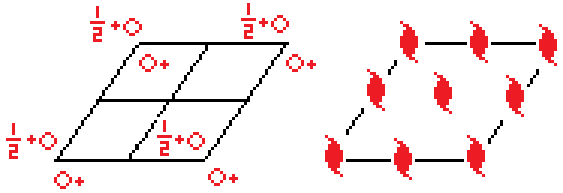


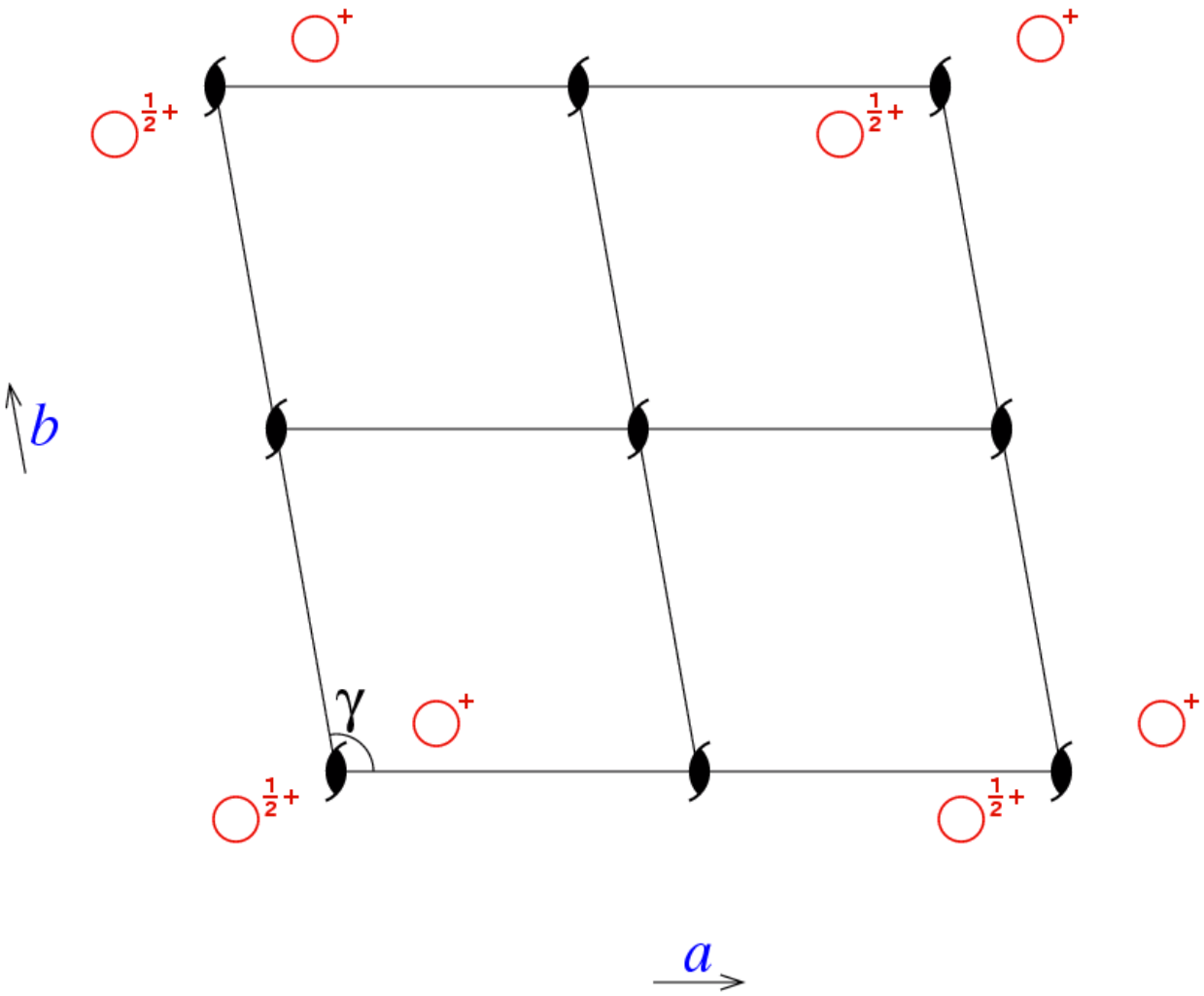
1 x, y, z

2 $\bar{x}, y, \bar{z}$

http://img.chem.ucl.ac.uk/sgp/large/003ay1.htm

Space group $P112_1$

$P2_1$ C_2^2		No. 4	$P112_1$	2 Monoclinic
				
Ist setting			Origin on 2_1; unique axis c	
Number of positions	Wyckoff notation	Point symmetry	Coordinates of equivalent positions	Condition limiting possible reflections
2	a	1	x, y, z; $\bar{x}$, $\bar{y}$, $\frac{1}{2}$ +z	General: hkl: No conditions hk0: No conditions 00l: l=2n

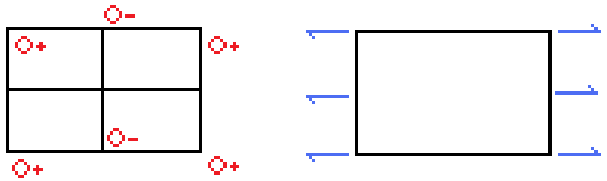


1 x, y, z

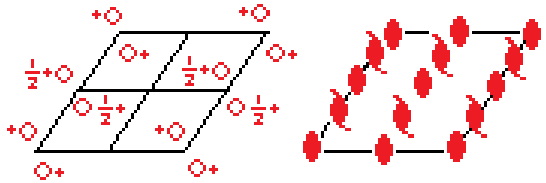
2 $\bar{x}, \bar{y}, \frac{1}{2} + z$

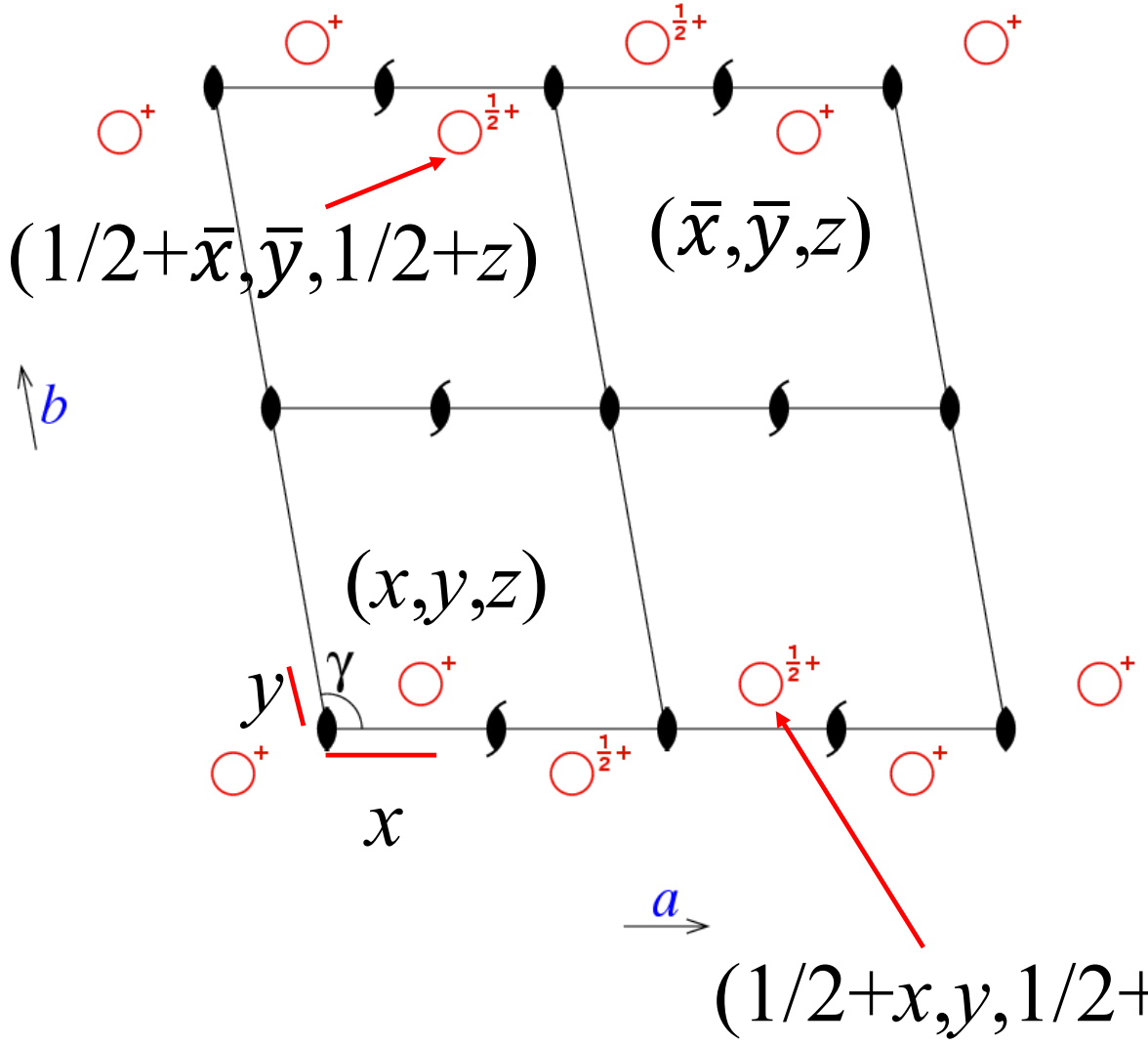
http://img.chem.ucl.ac.uk/sgp/large/004az1.htm

Space group $P12_11$

$P2_1$ C_2^2		No. 4	$P12_11$	2 Monoclinic
				
			Origin on 2_1; unique axis b	2nd setting
Number of positions	Wyckoff notation	Point symmetry	Coordinates of equivalent positions	Condition limiting possible reflections
2	a	1	x, y, z; $\bar{x}$, $\frac{1}{2}$+y, $\bar{z}$	General: hkl: No conditions h0l: No conditions 0k0: k=2n

Space group B112

B2 C_2^3		No. 5	B112	2 Monoclinic
				
Ist setting			Origin on 2; unique axis c	
Number of positions	Wyckoff notation	Point symmetry	Coordinates of equivalent positions	Condition limiting possible reflections
4	c	1	x, y, z; $\bar{x}$, $\bar{y}$, z $+(\frac{1}{2}\ 0\ \frac{1}{2})$	General: hkl: h+l=2n hk0: h=2n 00l: l=2n
2	b	2	0, $\frac{1}{2}$, z	Special: as above only
2	a	2	0, 0, z	



1 x, y, z

2 $\bar{x}, \bar{y}, z$

+ $(\frac{1}{2},0,\frac{1}{2})$

Here you can access the "**International Tables for Crystallography**"

http://it.iucr.org/

and all the space groups

http://it.iucr.org/Ab/contents/

What can we do with the space group information contained in the International Tables?

1. Generating a Crystal Structure from its Crystallographic Description

2. Determining a Crystal Structure from Symmetry & Composition

Example: Generating a Crystal Structure

http://chemistry.osu.edu/~woodward/ch754/sym_itc.htm

Description of crystal structure of Sr_2AlTaO_6

Space Group = $Fm\bar{3}m$; a = 7.80 Å

Atomic Positions

Atom	x	y	z
Sr	0.25	0.25	0.25
Al	0.0	0.0	0.0
Ta	0.5	0.5	0.5
O	0.25	0.0	0.0

From the space group tables

http://www.cryst.ehu.es/cgi-bin/cryst/programs/nph-wp-list?gnum=225

32	*f*	3m	*xxx*, -*x*-*xx*, -*xx*-*x*, *x*-*x*-*x*, *xx*-*x*, -*x*-*x*-*x*, *x*-*xx*, -*xxx*
24	*e*	4mm	*x*00, -*x*00, 0*x*0, 0-*x*0,00*x*, 00-*x*
24	*d*	mmm	0 ¼ ¼, 0 ¾ ¼, ¼ 0 ¼, ¼ 0 ¾, ¼ ¼ 0, ¾ ¼ 0
8	*c*	$\bar{4}$3m	¼ ¼ ¼ , ¼ ¼ ¾
4	*b*	m$\bar{3}$m	½ ½ ½
4	*a*	m$\bar{3}$m	000

Sr 8*c*; Al 4*a*; Ta 4*b*; O 24*e*

40 atoms in the unit cell

stoichiometry $Sr_8Al_4Ta_4O_{24} \rightarrow Sr_2AlTaO_6$

Another way of thinking: F lattice has for lattice sites per unit cell, each site have to put a molecule Sr_2AlTaO_6 (10 atoms) $\rightarrow$ 4x10 = 40 atoms per unit cell

F: face centered

$\rightarrow$ (000) (½ ½ 0) (½ 0 ½) (0 ½ ½)

Sr (000) (½½0) (½0½) (0½½)

8*c*: ¼ ¼ ¼ $\rightarrow$ (¼¼¼) (¾¾¼) (¾¼¾) (¼¾¾)

¼ ¼ ¾ $\rightarrow$ (¼¼¾) (¾¾¾) (¾¼¼) (¼¾¼)

Al ¾ + ½ = 5/4 =¼

4*a*: 0 0 0 $\rightarrow$ (000) (½ ½ 0) (½ 0 ½) (0 ½ ½)

Determining a Crystal Structure from Symmetry & Composition

Example:
Consider the following information:
Stoichiometry = $SrTiO_3$
Space Group = $Pm\bar{3}m$
a = 3.90 Å
Density = 5.1 g/cm^3

First step:
calculate the number of formula units per unit cell :
Formula Weight $SrTiO_3$ = 87.62 + 47.87 + 3× (16.00) = 183.49 g/mol (M)

Unit Cell Volume = $(3.90\times10^{-8}\ cm)^3 = 5.93 \times 10^{-23}\ cm^3$ (V)

$(5.1\ g/cm^3)\times(5.93 \times 10^{-23}\ cm^3)$: weight in a unit cell

$(183.49\ g/mole) / (6.022 \times10^{23}/mol)$: weight of one molecule of $SrTiO_3$

$\rightarrow (5.1\ g/cm^3) \times (5.93 \times 10^{-23}\ cm^3)/$
$(183.49\ g/mole/6.022 \times 10^{23}/mol) = 0.99$

$\rightarrow$ number of molecules per unit cell : 1 $SrTiO_3$.

From the space group tables (only part of it)

6	*e*	4mm	*x*00, -*x*00, 0*x*0, 0-*x*0,00*x*, 00-*x*
3	*d*	4/mmm	½ 0 0, 0 ½ 0, 0 0 ½
3	*c*	4/mmm	0 ½ ½ , ½ 0 ½ , ½ ½ 0
1	*b*	m$\bar{3}$m	½ ½ ½
1	*a*	m$\bar{3}$m	000

http://www.cryst.ehu.es/cgi-bin/cryst/programs/nph-wp-list?gnum=221

#2 CsCl structure

atoms	Number of positions	Wyckoff notation	Point symmetry	Coordinates of equivalent positions
Cl	1	a	m3m	0, 0, 0
Cs	1	b	m3m	$\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$

CsCl Vital Statistics	
Formula	CsCl
Crystal System	Cubic
Lattice Type	Primitive
Space Group	Pm3m, No. 221
Cell Parameters	a = 4.123 Å, Z=1
Atomic Positions	Cl: 0, 0, 0 Cs: 0.5, 0.5, 0.5 (can interchange if desired)
Density	3.99

#3 $BaTiO_3$ structure

atoms	Number of positions	Wyckoff notation	Point symmetry	Coordinates of equivalent positions
Ba	1	a	m3m	0, 0, 0
Ti	1	b	m3m	$\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$
O	3	c	4/mmm	0, $\frac{1}{2}, \frac{1}{2}$; $\frac{1}{2}$, 0, $\frac{1}{2}$; $\frac{1}{2}, \frac{1}{2}$, 0

Temperature

183 K 278 K 393 K

rhombohedral (R3m) Orthorhombic (Amm2) Tetragonal (P4mm). Cubic ($Pm\bar{3}m$)